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REVIEWS IN Aquaculture

REVIEW

Ice-Ice disease: An environmentally and microbiologically driven syndrome in tropical seaweed aquaculture

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Abstract

Seaweeds account for nearly 30% of global aquaculture production by volume, and their cultivation provides important opportunities for employment and wealth generation, particularly in rural coastal communities. Eucheumatoids (*Kappaphycus* and *Eucheuma* spp.) are tropical red algae commercially cultivated as raw material for production of carrageenans, particularly in South-East Asia and the Western Indian Ocean region. Diseases and pests are major limiting factors in eucheumatoid production, particularly a condition known as 'ice-ice' disease (IID). IID is characterized by bleaching of the thallus followed by the disintegration of affected tissues, leading to a loss of biomass and reduced carrageenan yield. IID has been associated with unfavourable changes in environmental conditions, particularly an increase in sea water temperature and a decrease in salinity, and infection by opportunistic or pathogenic bacteria. However, given that no single or combined causal agents have been definitely identified, IID may be considered a complex pathobiotic syndrome in which multiple factors contribute to the development of disease signs. Here, we review current knowledge of the abiotic and biotic factors associated with the development of IID, and the strategies employed to mitigate economic losses resulting from disease outbreaks. We discuss syndromic diseases in other marine holobionts, particularly other red algae and corals, in relation to their similarity to IID, and suggest the application of comprehensive and complementary methodologies, including high-throughput sequencing and high-quality microscopy, for enhancing our understanding of the pathological and microbiological processes associated with IID signs.

KEYWORDS
aquaculture, Eucheumatoids, Holobiont, Ice-Ice disease, Pathobiome

1 | INTRODUCTION

Diseases and pests are major limiting factors to aquaculture production,[1] including seaweeds,[2] prompting global action to improve biosecurity and knowledge of important yield-limiting pathogens affecting sustainable production, both now and in the future.[3,4] Seaweeds comprise almost 30% of global aquaculture production by volume.[5] Production of the carrageenophyte red algal genera *Kappaphycus* and *Eucheuma* (collectively known as eucheumatoids) has increased rapidly in the past decade, accounting for 33.99% of global aquatic plant cultivation in 2018.[5] Eucheumatoids are commercially cultivated for production of carrageenans, polysaccharides widely used in the pharmaceutical, cosmetic and food industries. Much of this cultivation occurs in South-East Asia, particularly

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Indonesia, the Philippines and Malaysia, with smaller scale production in the Western Indian Ocean (WIO) region including Kenya, Madagascar, Mozambique and Tanzania. It forms an important wealth generator, particularly in rural settings, with ~1 million farmers directly employed, and a greater number indirectly supported by, the sector in Indonesia alone.[6] On the coast of Tanzania and the Zanzibar archipelago, eucheumatoid cultivation indirectly supports 31,000 farmers, the majority of whom are women, with the proceeds contributing to an increase in living standards for themselves and their dependents.[7]

The expansion of eucheumatoid cultivation, particularly in regions outside the native geographical range of commercial species, has not been without problems. Coastal ecosystems located both directly at the site of cultivation and adjacent to farms may be affected.[8] The siting of eucheumatoid farms over seagrass beds, as is common practice in East Africa, not only leaves seagrasses vulnerable to trampling and shading, but has also been shown to alter seagrass bed community structure.[8,9] The invasive potential of eucheumatoid species is also well recognized. Fragments of escaped *Kappaphycus* from cultivation sites can colonize adjacent seagrass beds and reef patches, smothering native taxa.[10–12] *Kappaphycus* species originally introduced for cultivation trials in Kāne'ohe Bay, Hawaii, in the 1970s have spread rapidly from their sites of introduction and are now commonly found on coral reef slopes, where they overgrow and kill native corals and algae.[13]

Nevertheless, the sustainable expansion of seaweed aquaculture forms an important component of the aquaculture strategies of many nations, and being extractive in nature, it has significant potential as a low greenhouse gas (GHG) and low-impact aquaculture sector which may be integrated with other farming systems on land and in water.[14]

Eucheumatoid production is increasingly impacted by pests and diseases, with a condition known as 'ice-ice' disease (IID) posing a particular risk during cultivation. The condition was first observed in the Philippines in the 1970s[15,16] and is commonly encountered in eucheumatoid farms in all producing regions.[17,18] The term 'ice-ice' was coined by farmers in the Philippines using the English term 'ice' to describe the 'senescent tissue devoid of pigment that causes healthy branches to break off'.[19] While descriptions of IID in literature vary (Table 1), the condition is typically characterized by a whitening (loss of pigmentation) of affected thallus segments (Figure 1a–c), followed by a softening of the tissues. IID is marked by progressive disintegration of tissue and a loss of biomass (Figure 2), particularly if the disease occurs on the primary branches attaching individuals to cultivation ropes or lines, causing plants to detach.[20] This loss of biomass lowers farm yield (in some cases by up to 70%) and reduces the quality and quantity of the carrageenan produced by 25%–40% from affected seaweeds.[21,22] Lower quality carrageenan is less valuable, lowering the market value of affected crops.[23,24]

The products of eucheumatoid aquaculture are an important source of income for coastal communities in producing regions, and disease-related crop failure often drastically lowers annual income for participants.[25–27] In Indonesia, IID incidence can be very high (estimated 70%–100%[28,29]), leading to significant economic losses on local scales. An outbreak of IID in farms on Pari Island in the early 2000s caused damage to 60 tonnes of eucheumatoid crops, resulting in a financial loss of 169–180 million IDR ($17,300–18,500 USD).[28] More recent localized outbreaks in Nusa Penida Island, Bali and North Dullah Sub-Regency, Moluccas, resulted in similar losses in revenue.[30] Financial losses directly attributable to IID are less thoroughly documented in other producing countries, though value chain analysis suggests that loss of productivity as a result of IID is the primary problem for stakeholders in the Philippines,[31] and the Sabah Department of Fisheries in Malaysia consider IID as one of the most important challenges to the Malaysian seaweed industry.[32,33] These disease-associated losses in seaweed mirror broader recognition of disease as the key limiting factor in global animal aquaculture.[1]

2 | SIGNS OF DISEASE

Many reports of IID in the published literature come from local language journals and grey literature from major producing countries in South-East Asia (Table 1; those not subject to peer review marked with asterisks). Many lack a detailed description of the signs observed (Table 1). Where a description is given, the typical sign is whitening, depigmentation or 'bleaching' of the thallus (Figure 1a–c). IID bleaching can usually be distinguished from that caused by excessive UV exposure or desiccation, which occurs only at the tips of growing plants and is localized to those parts of the thallus which are above the surface of the water for prolonged periods of time[34] (Figure 1d). In contrast, IID depigmentation is most commonly observed at the base of the thallus but may also spread to apical regions (Figure 2a, b).[21,35,36] This depigmentation is often followed by the softening of affected thalli, and the eventual disintegration of diseased tissue[34,37,38] (Figure 2c).

Progression of IID is varied (Table 1), with no definitive studies to date attempting to document disease progression systematically in individual specimens. Hardening of affected tissue is infrequently reported and is also followed by breakage of affected thalli when observed.[39] There is also variation in descriptions of the appearance of affected tissues. Early reports describe 'powdery' growth on the thalli prior to the loss of pigmentation[40,41] which is not reported elsewhere. Similarly, some descriptions mention the appearance of a 'moist organic substance', 'slime' or 'mucus' on affected thalli,[34,39,42] which does not appear to be common. Importantly, an accurate and unified description for IID has not been produced to date.

3 | IID PATHOLOGY

Four studies have attempted to characterize microscopic changes in tissues affected by IID. In his PhD thesis, Aris[35] inoculated healthy *Kappaphycus alvarezii* with a purportedly pathogenic strain of *Vibrio alginolyticus*, resulting in 'chlorosis' (loss of pigment) and loss of

TABLE 1 Recorded incidence of 'ice-ice' disease (IID) in eucheumatoids (*Kappaphycus* and *Eucheuma* spp.)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
Kappaphycus alvarezii (var. brown; red; green)	Brazil	Strong water currents and winds	None given	Floating raft with tubular nets	NA	Field observation	178
K. alvarezii	Brazil	Decrease in water salinity and increase in water temperature	None given	Not stated	NA	NA	179
K. alvarezii	Brazil: Pitimbu Beach	Infection with filamentous algae (*Colaconema infestans*)	'Thallus depigmentation'	NA – drifting thalli originating from nearby cultivation sites	Yes	Laboratory manipulation of environmental conditions	65
E. spinosum (= *E. denticulatum*)	Caribbean: Guadeloupe	None given – only affected plants imported from the Philippines, not native individuals	None given	Experimental cultivation	NA	Field observation (experimental cultivation attempt)	180
K. alvarezii	Caribbean: Saint Lucia; Sain Martin; Grenadines	None given	None given	Various	NA	Field observation	152
Eucheuma denticulatum (var. red); *K. alvarezii* (var. red; green); *K. striatus* (var. green)	China: Hainan Province	Significant decrease in salinity and increase in water temperature	None given	*Eucheuma* and *Kappaphycus* co-cultured on floating rafts	NA	Field observation	47
K. alvarezii	China: Lian Bay, Hainan Province	Low salinity and shallow culture depth	None given	Square floating rafts at varying depths (50 – 150 cm)	NA	Field observation	181
K. alvarezii	Ecuador	None given	None given	Off-bottom monoline (in contained tanks)	NA	Field observation (experimental cultivation attempt)	152
E. striatum (=*K. striatus*); *E. spinosum* (= *E. denticulatum*)	Federated States of Micronesia: Pohnpei	Fungal infection (*Scopulariopsis brevicularis*)	'White powdery growth on the thallus which causes loss of pigments and disintegration of the alga'	Not stated	NA	Fungal culturing	41
K. alvarezii	Fiji: Rakiraki	'Unfavourable' environmental conditions	'Older portions of thalli frequently showed loss of pigmentation and degeneration'	Monoline	NA	Field observation	182

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii	India: Gulf of Mannar	High water temperature; Low salinity; Low water movement; Increase in bacterial diversity	'Standard symptoms of whitening of the branches and initiated with the colour changes of the thalli become transparent'	Monoline; floating rafts	Yes	Field observation; Bacterial culturing	183
K. alvarezii	India: Gulf of Mannar	Extreme changes in environmental conditions	Stressed seaweeds 'produce some moist organic substance that invites bacteria from the surrounding water and affects the host tissue, which in turn become white and hard'	Not recorded	No	Isolation and 16S sequencing of bacteria	39
K. alvarezii	India: Tamil Nadu	High water temperature; High light intensity; Low water movement	None given	Not recorded	No	Field observation; Bacterial culturing	74
K. alvarezii	India: Tamil Nadu	Carrageenan-degrading bacteria	'Characterised by "bleaching or whitening" white indications on the upper seaweed tissue"	Not stated	No	Histopathology; Spectroscopy; Gram staining; 16S Illumina sequencing	44
Eucheuma (assumed *E. denticulatum*)	Indonesia	'Alteration' in environmental parameters: water current, temperature and turbidity	'Appearance some bright colour spot in the thalli and will reduce seaweed colour until being white and broken-off easily'	NA	NA	NA (report of incidence only)	184
K. alvarezii	Indonesia: Pulau Panggang, Kepuluan Seribu,	Bacterial infection (*Flavobacterium*; *Vibrio*; *Pseudomonas* spp.; *Plesiomonas*)	'Symptoms of "ice-ice" disease are generally characterised by bleaching at the base of the thallus, middle and tip of the young thallus, which precedes the change in colour of the thallus to clear or transparent white' (translated from Indonesian)	Longline	Yes	Isolation and characterization of bacteria (biochemical tests; 16S sequencing); Pathogenicity assays; Histological examination	35 *

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii	Indonesia: Tolo Bay and Tomini Bay, Central Sulawesi	Infection with *Vibrio* spp.	'Causes plants to develop white spots on their thallus' (translated from Indonesian)	Not stated	No	Field observation; Bacterial isolation and biochemical characterization	63
K. alvarezii	Indonesia	Sudden changes in salinity, water temperature and light intensity. Wounds from grazers followed by secondary bacterial infection	'Characterised by red spots on part of the thallus, which gradually turn pale yellow and eventually turn white. The thallus becomes brittle and breaks easily' (Translated from Indonesian)	Rafts	NA	Field observation	86
K. alvarezii (var. green)	Indonesia: Lampung, Sumatera Island	Bacterial infection, and induced by 'drastic changes' in water quality	None given	NA	NA	NA	185
K. alvarezii	Indonesia: South Sulawesi	Bacterial infection (*Stenotrophomonas*; *Ochrobactrum*; *Pseudomonas*)	'Bleaching or whitening on upper tissue of the seaweed'	Not stated	Yes	Bacterial isolation and biochemical characterization	186*
E. spinosum (= *E. denticulatum*)	Indonesia: Kutuk Kecamatan, Denpasar Selatan, Kabupaten Badung	High water temperature and decrease in salinity due to runoffs and freshwater discharge; Infection with *Vibrio alginolyticus* and *Pseudomonas aeruginosa*	'Bleaching at the base of the thallus, middle and tip of the young thallus, which begins with a change in colour of the thallus to clear or transparent white' (translated from Indonesian)	Not stated	No	Bacterial isolation; morphological and biochemical characterization	36
K. alvarezii	Indonesia: South Sulawesi	Attachment of epiphytes	'the affected part is white and very fragile, if the disease attacks at the base of a binding seaweed, it can cause the release of the rope stretch'	Not stated	NA	Field observation	20
K. alvarezii	Indonesia: Karimunjawa Island	Bacterial infection (*Alteromonas*; *Pseudoalteromonas*; *Aurantiomonas*)	None given	Not stated	No	Bacterial isolation and 16S sequencing	78

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
E. cottonii (= K. alvarezii)	Indonesia	Change in environmental conditions linked to seasonal change from dry to wet season	'Whitened segments in nearly all of the seaweed branches'	NA	NA	NA	187
Kappaphycus sp. (var. brown and var. green)	Indonesia	Infection with presumed *Vibrio* spp.	Colour change, followed by 'release of mucus' by infected thalli, then the development of white spots and thallus fracture	Not stated	No	Bacterial isolation and biochemical characterization	42
K. alvarezii	Indonesia: Vid Bangir Bay, Tual, Moluccas	Nutrient depletion and high water turbidity	'Characterised by the appearance of spots on part of the thallus which gradually turn pale yellow and eventually turns white and breaks easily' (translated from Indonesian)	Longlines	No	NA	56
Kappaphycus spp.; *E. denticulatum*	Indonesia: Rote Island	Abrupt changes in water temperature and/or salinity. Potential association with epiphytic algae	'Loss of pigmentation of the algal thallus until it becomes white, followed by dissolution of the tissue until the two parts separate and a portion of the alga is lost'	NA	NA	NA	38
K. alvarezii	Indonesia: Kupang, East Nusa	Bacterial infection	'Changes in the thallus to become rough, wilted and wrinkled followed by the appearance of white spots on the surface of the thallus'	Not stated	Yes	Incubation of healthy seaweeds with diseased tissues; histological examination	43
K. alvarezii	Indonesia	Presumptive pathogenic bacterial infection	Started with the loss of mucous at the surface of thallus which led to a rough surface and subsequently a colour change from bright to pale. White spots/patches are later observed on the apex of the thallus which then spreads to the entire thallus	Long line	NA	Field observation, water quality testing	82

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii	Indonesia	Non-favourable biotic and abiotic environmental parameters (temperature, salinity, light intensity and extreme (sic) microbe populations) which led to a stressful condition which then reduces the immunity and increases disease susceptibility to the seaweeds	Red spots or patches on parts of the thallus which then turns pale yellow and subsequently white and soft due to necrosis	NA	NA	Water quality testing, bacteria isolation and biochemical characterization	80
E. denticulatum; *K. alvarezii*	Japan: Tosa Bay	Low water salinity	None given	Floating culture; laboratory tanks	NA	Laboratory induction of IID by manipulation of growth conditions	54
K. alvarezii	Japan: Tosa Bay	Proliferation of bacteria	'Whitened tips or segments'	Floating culture	Yes	Direct enumeration of bacteria from homogenized tissue by epifluorescence	50
K. alvarezii	Japan: Kōchi Prefecture (laboratory)	Influence of abiotic factors (water movement and temperature) on infection by *Vibrio* and *Cytophaga*	None given	NA (laboratory tanks)	Yes	Laboratory infection of abiotically stressed individuals with bacterial inoculations	57
E. denticulatum (var. brown); *K. alvarezii* (var. green, var. brown)	Kenya	High water temperature; High water salinity; low water motion; Damage by grazers	None given	Fixed off-bottom	NA	Field observation	60
E. denticulatum; *K. alvarezii*	Kenya: Kibuyuni and Mkwiro	None given	'White and soft thallus'	Modified off-bottom Floating raft	NA	Comparison of the prevalence of diseases and pests (including IID) with different culture methodologies	188

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
E. denticulatum; *K. striatus*	Madagascar	High water temperature; Low water nitrogen	None given	Monoline	NA	Field observation	58
K. alvarezii	Malaysia: Sabah, Borneo (farmed and laboratory culture)	IID signs induced by elevated growth temperatures are enhanced by inoculation with *Vibrio* sp. Each factor was also able to induce IID signs in the absence of the other	'Characterised by a change in seaweed pigment and softening of the thallus … this decreases the carrageenan yield and produces inferior quality carrageenan'	NA (individuals maintained in laboratory conditions)	Yes	Laboratory induction of IID signs Bacterial isolation and biochemical characterization, 16S sequencing	37
E. isiforme (= *Eucheumatopsis isiformis*)	Mexico: Yucatán	None given	'Whitening and dissolution of the plant'	Individuals obtained from wild populations were cultured on rafts with monofilament lines (depth 1.5m)	NA	NA	49
K. alvarezii	Mexico: Yucatán	IID most frequently observed when seawater temperatures exceeded 30 °C	None given	Off-bottom	NA	Field observation (experimental cultivation attempt)	152
K. alvarezii	Panama	None given	None given	Floating rafts	NA	Field observation (experimental cultivation attempt)	152
E. spinosum (= *E. denticulatum*); *K. striatus*	Philippines: Hingutanan Island, Bohol	High water phosphate concentration; epiphyte infestation; Secondary bacterial infection	'A white powdery growth on the thallus which causes the loss of pigments, gradual consumption and fragmentation of the alga'	Not stated	Yes	Field observation; Monitoring of environmental parameters; Bacterial isolation and biochemical characterization	40

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii (var. brown)	Philippines	High light intensity and calm weather *low water motion); also appeared related to nitrogen limitation, particularly during periods of low water motion	'white soft portions usually on the main branches of the thallus. The portions with "ice-ice" may become colonized by microorganisms which appears as distinct white, pinkish or reddish patches of slime on the infected parts. The portions thus infected softens further and dissolved away resulting to the breakage of the branches'	Bottom monoline Raft	NA	Field observation	34
K. alvarezii; *E. denticulatum*	Philippines: Danajon Reefs, Bohol	Bacterial infection (*Vibrio* and *Cytophaga*)	None given	Not stated	Yes	Bacterial isolation and biochemical characterization Laboratory induction of IID signs by inoculation with bacterial isolates	76
K. striatus	Philippines: Calatagan, Batangas	None given	'Whitening and softening of the infected part, usually the base of the thallus … leads to fragmentation and eventual loss of biomass'	Not stated	Yes	NA	21
K. striatus (var. sacol); *K. alvarezii* (var. tambalang, var. giant kapilaran, var. orang giant); *E. denticulatum* (var. spinosum)	Philippines: Batangas; Bohol; Zamboanga	Culturable carrageenase-producing bacteria (presumptive *Vibrio* spp.)	'Thallus whitening followed by softening, the most obvious manifestation of the disease, is indicative of the involvement of carrageenase-producing bacteria'	Fixed bottom monoline Floating raft	Yes	Bacterial isolation and biochemical characterization	75

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii (var. brown and var/ green); *E. denticulatum* (var. brown and var. green)	Philippines: Negros Oriental, Zamboanga del Norte	High water temperature and low tides, low salinity, water pollution from factory discharge High incidence of epiphytes	'When a seaweed is under stress it emits a moist organic substance that attracts bacteria in the water and induces the "whitening" and hardening of the seaweed branches. Uninfected parts remain healthy while infected ones undergo depigmentation and eventually lead to plant breakage by any force of nature'	Monoline Floating raft (offshore) Off-bottom monoline (offshore)	NA	Field observation	48*
K. alvarezii; *K. striatus* (var. green and var. orange)	Philippines: Calatagan, Batangas	Marine-derived fungi (*Aspergillus* and *Phoma*)	'Whitening and softening of the algal thallus, which eventually lead to low yield and poor quality carrageenan'	Not stated	Yes	Induction of IID signs in non-axenic cultures by inoculation with fungal isolates	127
K. alvarezii	Philippines	Wounds caused by mechanical injury (from water movement, grazers or epiphyte attachment) allow the proliferation and spread of bacteria	'One or more branches had thalli that were whitened and soft'	Bamboo raft with net cage	NA	Field observation	62
K. alvarezii	Philippines: Zamboanga Peninsula	High water temperature and pH; low water salinity and low water depth	'whitening of the branches and initiated with the colour changes of thalli which appeared transparent'	Fixed off-bottom	NA	Field observation	55
E. denticulatum; *K. alvarezii*	Tanzania: Zanzibar	Limited availability of mineral nutrients	None given	NA	NA	NA	72
K. alvarezii	Tanzania: Songo Songo	Stress induced by rapid fluctuations in environmental conditions	None given	Off-bottom (active and abandoned cultivation sites)	NA	Field observation	84
K. alvarezii	Venezuela	None given	None given	Various	NA	Field observation (experimental cultivation attempt)	152

(Continues)

TABLE 1 (Continued)

Affected species	Location	Attributed causes	Description of 'Ice-Ice' signs	Farming method	Healthy tissue examined?	Methodologies applied	Reference
K. alvarezii; *K. striatus*; *E. denticulatum*	Vietnam: Cam Ranh Bay; Van Phong Bay	Presumptive bacterial infection	'the bleaching and formation of white spots on the thallus'	Not stated	Yes	16S Illumina sequencing; water quality testing	79
K. alvarezii	Solomon Islands; Vanuatu; Kiribati	Environmental stress triggers the production of volatile hydrocarbons, resulting in necrosis and secondary bacterial infection	'Tissues are bleached and become necrotic'	NA	NA	NA	189*
K. alvarezii	Malaysia; Philippines; Indonesia; Tanzania	Epiphyte outbreaks lead to IID 'bacterial' disease	None given	NA	NA	NA	29

Notes: Descriptions of IID signs from publications in languages other than English have been translated (original language in parentheses). References which have not been subjected to peer review are marked with an asterisk. Field observation refers to studies in which causative factors were identified *in situ*, usually through monitoring of environmental parameters. Where no methodologies are listed, the cited study records the incidence of IID in eucheumatoids in the specified location.

biomass characteristic of IID. Histology of healthy tissue showed intact, compact and orderly arranged cells in the cortical and medullary layers. In diseased tissues, these structures were disrupted, cells were lysed, and carrageenan deposits were degraded. This breakdown of tissue was positively correlated with necrosis or cell death. In a later study, the same author observed progressive cell degradation associated with tissue damage or cell lysis in *K. alvarezii* affected by IID.[43] Similarly, Riyaz et al.[44] observed cell wall rupture and damage to subcortical cells of IID-affected tissue in comparison with healthy *K. alvarezii* tissue. Maulani et al.[45] also noted changes in the shape of cells and damage to the outer layers of tissue in IID-affected *K. alvarezii* collected in the field. The cortex of diseased tissue was irregular and did not form the normal round or oval shape of healthy thalli.

4 | SUSCEPTIBILITY OF EUCHEUMATOID TAXA TO IID

IID is known to affect all commercially exploited eucheumatoid species, including *Kappaphycus alvarezii*, *K. striatus*, *K. malesianus* and *Eucheuma denticulatum* (Table 1). All cultivated strains are thought to be affected, with apparently varying severity.[46] There is some evidence to suggest *E. denticulatum* is less susceptible to severe disease than *Kappaphycus*.[47] There are suggestions in the grey literature that this difference may relate to differences in the carrageenan types produced by each species, or differences in the texture of the thalli, making *Kappaphycus* more susceptible to colonization by microorganisms,[48] though this has not been demonstrated in peer-reviewed studies. IID has also been reported in the related species *Eucheumatopsis isiformis* (previously *Eucheuma isiforme*) at two locations in Yucatán, Mexico, during attempts at cultivation.[49] So far, there have been no reports of IID affecting wild eucheumatoids (either wild specimens of cultivated strains, or species which are not commercially exploited), though no studies explicitly state they have attempted to find IID outside of cultivation settings.

IID occurs in all regions where eucheumatoids are commercially produced, including the native range of commercial cultivars in Southeast Asia, and areas where these cultivars have been introduced for the purposes of cultivation in Africa and Latin America (Table 1). The majority of reported outbreaks are in Southeast Asia, affecting *K. alvarezii*, reflecting not only the higher productivity of this region, but also the greater dedicated research effort in comparison with the WIO region and areas with lower or sporadic production such as South America, India and the Pacific Islands.[17,19]

There are a small number of publications reporting similar disease signs as IID in cultivated *Gracilaria* spp. in Indonesia and the Philippines,[50–52] though none from major *Gracilaria* producing regions of China and Chile. This may suggest that IID occurs rarely in this genus, or that the term has been misapplied to macroscopically similar syndromes. For the purposes of this review, we discuss only the proposed causes of IID in eucheumatoids, and do not include observations of similar pathologies in *Gracilaria*.

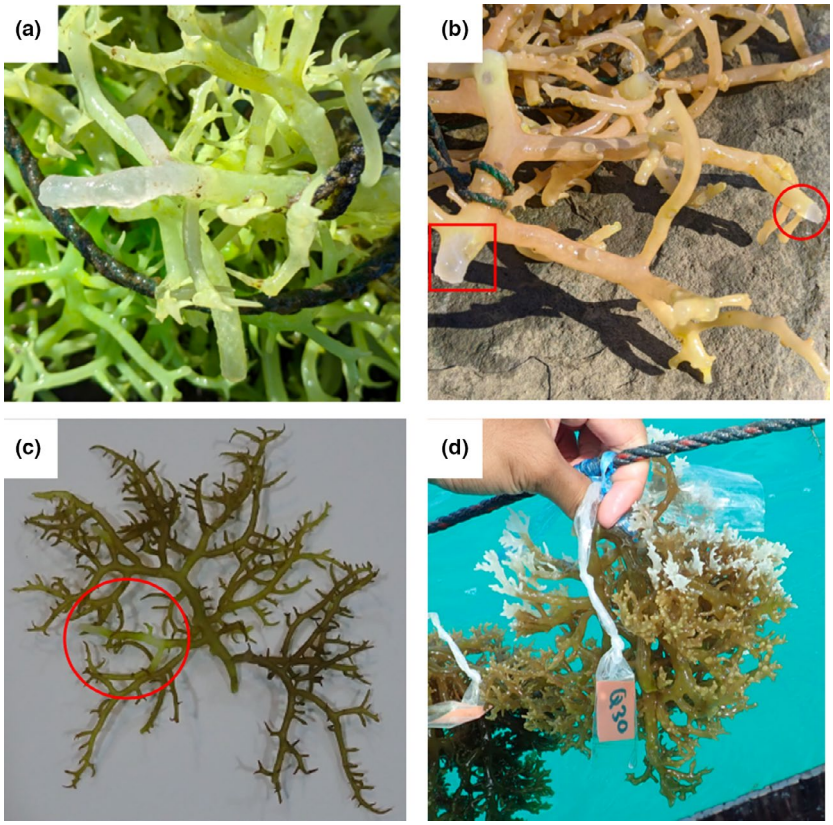


FIGURE 1 *Kappaphycus* spp. cultivated in Southeast Asia showing typical signs of 'ice-ice' disease (IID) bleaching affecting the secondary branch of *K. striatus* cultivar'*sacol*'(a), the primary branch (red square) and apex (circle) of *K. alvarezii* '*tambalang*' (b), and the secondary branch (red circle) of *K. malesianus* (c). In contrast, (d) shows UV-induced bleaching at the growing tips of *K. alvarezii*

5 | IID AND ENVIRONMENTAL STRESSORS

Occurrence of IID on eucheumatoid farms is highest during the hot-dry season in Tanzania (February–March), Indonesia (March–April) and the Philippines (March–May), which is largely attributed to higher water temperatures in several technical reports on eucheumatoid farming.[18,53] IID is frequently associated with 'unfavourable' changes in environmental parameters, in both field observation and laboratory studies (Table 1). Most often these include a decrease in salinity, coupled with higher water temperature.[47,54,55] Other environmental stressors, including low light levels,[54,56] slow water movement[34,57] and high water pH,[55] have also been associated with IID, as have water pollution[36] and low nitrogen availability.[34,58] Technical reports also suggest deposits of silt or terrigenous material on seaweed thalli may also induce IID bleaching by decreasing light absorption and leaving individuals more prone to bacterial colonization.[59]

Mechanical injury to seaweed thalli, often wounds caused by grazing by siganids (rabbitfish), turtles and microbenthic invertebrates[34,60,61] or as a result of seedlings being tied too tightly to cultivation lines, may render individual seaweeds more susceptible to IID.[62] These wounds may be more susceptible to microbial colonization and may present a route of entry and followed by proliferation of opportunistic pathogenic bacteria.[62,63]

Signs of IID frequently co-occur with epiphyte infestation,[38,64,65] and 'unfavourable' fluctuations in environmental conditions are considered predisposing factors to both conditions.[66,67] Epiphytic filamentous algae, particularly *Melanothamnus* spp., penetrate the cortical layer of eucheumatoid hosts, resulting in tissue damage and bacterial infection.[29,68]

Mtolera et al.[69,70] found that production of hydrogen peroxide and volatile hydrocarbons (VHCs) was higher in *Eucheuma denticulatum* subjected to environmental and mechanical stressors and was linked to both IID depigmentation and epiphyte infestation.[71,72] Extended exposure to the reactive oxygen species produced by this stress response may result in physiological damage and the thallus depigmentation characteristic of IID.[73]

6 | THE MICROBIOLOGY OF IID

The bacterial genera *Vibrio* and *Pseudomonas*, identified using biochemical tests, are frequently isolated from IID-affected eucheumatoid tissues (Table 2); however, many cultivation-based studies of IID use selective or differential media for *Vibrio* identification.[35,36,42,74,75]

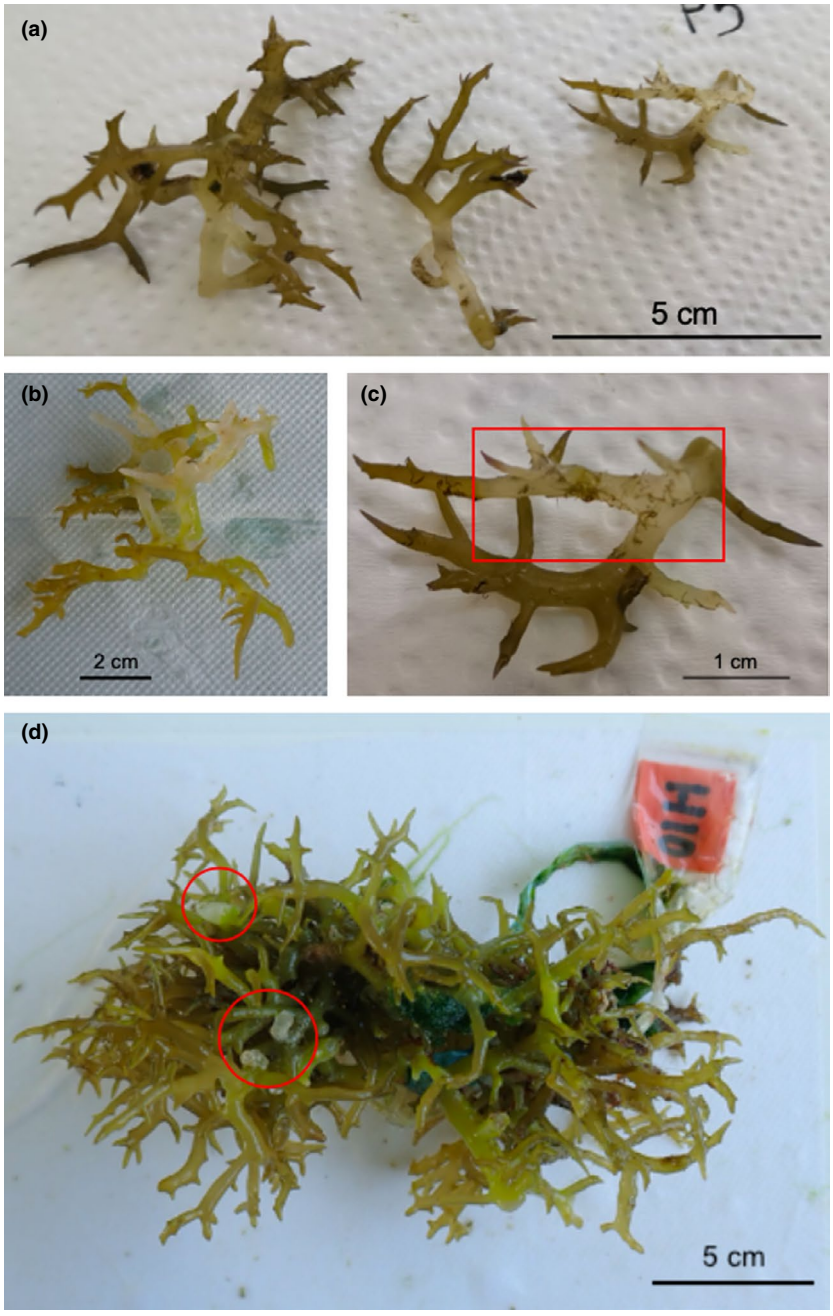


FIGURE 2 Cultivated *Kappaphycus* from Sabah, Malaysia, showing signs of 'ice-ice' disease (IID). (a) IID depigmentation of secondary branches and apices, (b) Significant IID infection affecting primary, secondary and apical tissues. (c) IID co-occurring with filamentous red algae (red rectangle), (d) Fouling biofilms affecting an individual also showing signs of IID

Several other bacterial genera have also been associated with signs of IID, particularly Gammaproteobacteria (*Pseudoateromonas*, *Alteromonas*, *Stenotrophomonas*) and Alphaproteobacteria (*Aurantimonas*, *Ochrobactrum*) (Table 2). In many studies, almost all bacteria isolated from IID-affected tissue are Gram negative, though Azizi et al.[37] acknowledge that this may potentially be the result of bias in the culturing conditions used. Largo et al.[76] report differences in the counts and diversity of colonies obtained using growth media, further suggesting that cultivation conditions influence bacteriological studies.

Few of the bacterial taxa isolated in these studies are fully characterized: identity is usually assigned on the basis of the result of a small battery of biochemical tests (Gram staining, catalase and motility tests and colony morphology are the most widely utilized), enabling taxonomic assignment to genus level at best. However, there is a lack of consistency in the application of these tests, and in some cases, the results of individual tests are not fully reported, hindering our ability to draw comparisons between the biochemical capabilities of isolated (presumptive) *Vibrio* strains.

TABLE 2 Summary of bacterial genera isolated from 'ice-ice' disease (IID) affected eucheumatoid tissue in cultivation-based experiments. Growth media marked with asterisks indicate selective differential media for *Vibrio*.

Isolate source	Growth medium	Total strains isolated	IID-associated genera	Isolates sequenced?	Healthy tissue examined?	Ability to induce IID	Reference
Eucheuma spinosum (= *E. denticulatum*)	Not stated	Not reported	*Pseudomonas* *Flavobacterium* *Vibrio* *Xanthomonas* *Achromobacter*	No	No	Not attempted	40
E. spinosum (= *E. denticulatum*)	Tryptic soy agar (TSA) Thiosulphate-citrate-bile salts-sucrose agar (TCBS)†	Not stated	*Pseudomonas* *Vibrio*	No	No	Not attempted	36
Kappaphycus alvarezii; *E. denticulatum*	FeTY agar	75 (only 10 included in pathogenicity tests)	*Vibrio* *Cytophaga* *Aeromonas* *Pseudomonas-Alteromonas*	No	Yes	Inoculation with *Vibrio* and *Cytophaga* strains induced development of IID; inoculation with other isolates did not	76
K. alvarezii; *K. striatus*; *E. denticulatum*	Marine Agar (MA) TCBS† FePY carrageenan agar	97 (diseased) 68 (healthy)	Presumptive *Vibrio* spp. Presumptive *Cytophaga* Uncharacterized unpigmented Uncharacterized pigmented	No	No	Not attempted	75
K. alvarezii	Seawater complex (SWC) TCBS†	5	*Vibrio* *Flavobacterium* *Pseudomonas* spp. *Plesiomonas*	Yes (*Vibrio* only)	Yes	*Vibrio alginolyticus* strain PNGK induced IID within 1 day post inoculation	35
K. alvarezii	Not stated	5 (diseased)	*Vibrio* spp.	No	No	Not attempted	63
K. alvarezii	Zobell marine agar (ZMA)	9 (diseased)	*Alteromonas* *Bacillus* *Pseudomonas* *Pseudoalteromonas* *Aurantimonas* *Rhodococcus*	Yes	No	Inoculation within *Alteromonas*, *Aurantimonas* and *Pseudoalteromonas* induced IID signs within 3 days. No other isolates induced IID	78

(Continues)

TABLE 2 (Continued)

Isolate source	Growth medium	Total strains isolated	IID-associated genera	Isolates sequenced?	Healthy tissue examined?	Ability to induce IID	Reference
K. alvarezii	ZMA	Not stated	Not characterized	No	No	Not attempted	183
K. alvarezii	Bacto agar (BA) Plant agar (PA) Marine agar (MA)	40 (diseased) 12 (healthy)	Farmed: *Vibrio* spp. *Halomonas* *Aestuariibacter* *Alteromonas* Tissue cultured: *Primorskyibacter* *Pseudoalteromonas* *Thalassospira*	Yes (abundant isolates only)	Yes	*Vibrio alginolyticus* strain ABI-TU15 was shown to induce IID signs in both farmed and tissue cultured *K. alvarezii*, with signs more severe in farmed individuals. No other strains were tested for the ability to induce IID	37
K. alvarezii	ZMA TCBS†	Not stated	*Bacillus* *Vibrio*	No	No	Not attempted	74
K. alvarezii	ZMA	83 (diseased; 6 abundant) 61 (healthy)	*Pseudoalteromonas* *Planococcus* spp. *Bacillus* spp. *Brevibacillus*	Yes (abundant isolates only)	Yes	Not attempted	39
K. alvarezii	Alkaline Peptone Agar (APA)	Not reported	*Bacillus*, *Vibrio*, *Pseudomonas*, *Flavobacterium*	No	No	Not attempted	80
Kappaphycus sp. (vars red and green)	TCBS†	Not reported	*Vibrio*	No	No	*Vibrio* isolates induced IID in both cultivars after 4 days (green) and 7 days (red)	42

Aris (2011) (PhD thesis), Azizi et al. (2018) and Largo et al. (1995b) included agar hydrolysis in the tests utilized in their studies of bacteria associated with IID in *Kappaphycus alvarezii* and showed that the *Vibrio* strains isolated in each case were capable of degrading agar; 16S rRNA data are available for the strains isolated by Aris (2011) and Azizi et al. (2018), but not for that of Largo et al. (1995b).[35,37,76] These sequences show low identity (92%) to each other, and thus do not represent the same species or strain. Each lineage was able to induce IID in inoculated healthy tissue, which may suggest that IID-associated *Vibrio* spp. are opportunistic, rather than being host-specific, and possibly taxonomically diverse. Searches of both sequences against the NCBI GenBank database reveal that the sequence of Aris (2011)[35] is not identical to any characterized species, instead showing equal sequence identity (98.4%) to characterized *Vibrio* species *V. alginolyticus*, *V. diabolicus*, *V. neonatus*, *V. orientalis* and *V. parahaemolyticus*. The much shorter (130 bp compared with 1300 bp) sequence of Azizi et al.[37] is 100% identical to *V. alginolyticus* and *V. diazotrophicus* (a non-pathogenic, nitrogen-fixing species). The 16S rRNA gene is a particularly poor taxonomic marker for the delineation of *Vibrio*,[77] and so the identity and phylogenetic relationships of the agar-degrading strains isolated in these studies remain undetermined.

Only two other studies have attempted 16S sequencing for the molecular characterization of taxa isolated from IID-affected seaweeds.[39,78] In most cases, the seaweed-derived bacterial sequences are 95–98% identical to the closest characterized matches in publicly available sequence databases and therefore likely not the same species. As such, the exact identity of these isolates remains undetermined.

The lack of studies comparing microbiota associated with healthy and diseased tissue makes the role of bacteria in the development of IID difficult to interpret. It is unclear in most cases whether the bacterial taxa implicated are part of the normal microbiota of their host macroalgae or opportunistically colonize compromised tissues from the surrounding environment. Of the handful of studies that do characterize bacteria associated with healthy tissue from the same host, Butardo et al.[75] reported differences in the characteristics of bacterial isolates from IID-afflicted and healthy *K. alvarezii* tissues, noting agarolytic and κ-carrageenanolytic activity was only observed in strains isolated from diseased tissue. Kopprio et al.[79] found OTUs identified as *Vibrio* and *Alteromonas* associated with all eucheumatoid tissues, however noted these OTUs differed between healthy and diseased states, which may suggest successive colonization by different bacterial strains as IID progresses.

7 | CURRENT MANAGEMENT PRACTICES

Few evidence-based, reliable strategies for preventing outbreaks of IID have been developed, and most mitigation measures are focused on minimizing associated economic losses. These measures fall into two broad categories: those aiming to reduce the occurrence of IID outbreaks, and those aiming to reduce the impact and spread of IID.

Most practices designed to reduce the occurrence of IID aim to limit the exposure of seaweeds to associated environmental stressors.[71] This is most easily achieved through the choice of farming method. In Malaysia, farmers most frequently opt for longline cultivation in water several metres deep. This reduces exposure to temperature fluctuation and to air during low tides when compared to other methods. Cultivation of bacteria from water collected at inactive eucheumatoid cultivation sites with a history of IID incidence found that the water closer to the surface (depth 20–50 cm) contained carrageenan-hydrolysing bacteria (*Bacillus*, *Vibrio)* and potential opportunistic pathogens (*Flavobacterium*, *Pseudomonas*), which were not found in deeper water,[80] suggesting siting eucheumatoid farms in deeper water may limit exposure to potentially disease-influencing bacteria.

Comparison of longline and off-bottom methods for the cultivation of *Kappaphycus alvarezii* in Madagascar have shown disease incidence to be lower using longline techniques.[81] Aris and Ibrahim[82] compared the horizontal transmission of IID at various stocking densities of *K. alvarezii* (20, 30 and 40 cm distance between seedlings) and found that loss of biomass as a result of IID over time is inversely proportional to the distance of the seedlings. Farmers may also favour lower stocking densities as this may enhance the growth of *Kappaphycus alvarezii* and *E. denticulatum*.[83]

A recent survey of disease incidence in eucheumatoid farms employing a range of cultivation techniques across the Philippines found no evidence of a link between farming technique or water depth and the observation of IID signs, suggesting mitigation of IID incidence by manipulation of cultivation conditions is not straightforward.[66]

Farmers regularly monitor their crops for biofoulers, usually loosely attached algae on the thalli and cultivation ropes,[84] or fouling biofilms, and remove these in order to reduce their negative effects. They may also choose to plant preferentially during seasons with more favourable environmental conditions.[18,56,85] Finally, in some cases farmers choose crops which show higher disease resistance: in areas where *K. alvarezii* is particularly difficult to cultivate due to high incidence of IID, many farmers are switching to *Eucheuma denticulatum*, which appears to be more resistant to most diseases and pests, including IID.[86] *E. denticulatum* var. *million-million* was recently introduced to cultivation sites in Tanzania from the Philippines, where it shows greater resilience to epiphytes and IID than previously used strains.[18] In his PhD thesis, Nor[85] states that farmers in Malaysia may choose different varieties of *K. alvarezii* depending on seasonality and seedling availability, though in most cases farmers prefer to use strains which in their experience have grown well previously at their sites (pers comm. Malaysian farmers). Between seasons, farmers are advised to carefully clean culture equipment such as ropes, particularly following IID and epiphyte outbreaks[87]; however, similar measures have only been partially effective in preventing disease in other cultivated algae.[88]

IID is thought to 'spread' between plants when individuals are grown closely together on culture lines.[28,43,62,82] Cultivation conditions, particularly the density at which seedlings are planted, will also influence abiotic parameters, particularly lowering nutrient and light

availability due to crowding, and reducing water movement around plants. The most common strategy for reducing the potential spread of IID, both within affected individuals and across farms, is the regular close monitoring of crops and removal of diseased portions of the thallus.[17] More severe IID outbreaks are dealt with by early harvesting at affected sites, similar to the 'emergency harvesting' that occurs in other aquaculture sectors such as shrimp.[1] In Indonesia, it is recommended to farmers to harvest crops early if an outbreak persists for 10 days or more,[89] with similar practices employed in Malaysia and the Philippines.[17]

8 | IID AS A SYNDROMIC DISEASE

The pathological processes behind the development of IID signs remain relatively unknown. IID is often stated to be the result of complex interactions between abiotic stress induced by 'unfavourable' environmental conditions and opportunistic colonization by 'pathogenic' bacteria.[66,67] As such, IID fulfils the criteria for definition as a syndrome (e.g. a recognizable complex of signs and other physical findings which indicate a specific condition for which a direct cause is not necessarily understood[90]). Syndromic diseases are common in marine systems and can have significant ecological and economic impacts. Collectively, the so-called 'white' syndromes of corals[91]—a group of macroscopically similar signs of tissue loss of unknown aetiology affecting a wide number of species—have contributed to the large-scale mortality of corals in the Florida Reef Tract (Stony Coral Tissue Loss Disease[92]), the Caribbean (White Band Disease[93]) and Great Barrier Reef, Australia (various coral bleaching events[94]). White faeces syndrome (WFS) of shrimp *Litopenaeus vannamei*, characterized by white faecal strings, distended midgut and white-to-golden faeces, is responsible for significant losses in cultivation settings in Asia.[95,96] The exact causes of WFS remain undetermined, though the condition has been linked to pathogenic strains of *Vibrio*,[97] changes in gut prokaryote community structure and function,[95] and the presence of opportunistic pathogenic bacteria in the water column of cultivation ponds.[96] The condition is increasingly linked to infection with the microsporidian *Enterocytozoon hepatopenaei* (EHP),[98] although other studies suggest that EHP is not causative, with the parasite thought to exacerbate underlying causes of WFS.[99] There is limited evidence to suggest that the biogeochemical conditions in cultivation ponds, particularly water pH, temperature and turbidity, may be linked to the development of WFS disease, and therefore serve as indicators for disease risk.[96]

9 | LIMITATIONS OF RESEARCH TO DATE

Despite the general consensus that IID signs develop in response to a number of factors, often in combination, most studies have focused on the impact of a single factor, either an environmental stressor or a purported pathogen. Only two studies have attempted to characterize the interactions of multiple stressors. Both Largo et al. (1995b) and Azizi et al. (2018) explored the relationship between IID-promoting bacteria and environmental stressors (low salinity and elevated growth temperature, respectively).[37,76] Both studies showed that while *Vibrio* strains P11 and ABI-TU15 are capable of inducing IID *in vitro* in the apparent absence of other reported causative factors, in the presence of environmental stressors these same strains can rapidly induce more severe IID signs.

It should, however, be noted that observations *in vitro* do not necessarily reflect natural processes. Two bacterial strains, *Nautella italica* R11 and *Phaeobacter gallaeciensis* LSS9, are able to induce bleaching of the red alga *Delisea pulchra* under laboratory conditions[100,101] and have both been shown by high-throughput sequencing studies to be present in diseased populations in natural settings. However, Zozaya-Valdes et al.[102] showed only strain LSS9 was statistically significantly more abundant in diseased tissues in the field, indicating that *in vitro* experiments often simplify *in vivo* interactions.

Research to date has been biased towards identifying causative factors, with little effort given to characterizing structural or functional changes in tissues showing signs of IID. Gross pathology (macroscopic signs) can often be very similar for what are microscopically very different processes with different causative factors (as noted by Largo 2002). There is some inconsistency in the signs attributed to IID, with a number of reports including uncommon signs such as hardening of affected tissues, or the presence of either powdery substances or mucus.[39,40,42] It remains to be determined whether the gross pathology observed in all cases is the result of the same causative factor(s), or whether IID defines a collection of separate conditions which manifest with similar gross signs, but with different aetiologies (the 'moving target' hypothesis[103]). There is very little available literature for 'normal' or healthy tissue structure in eucheumatoids, and limited light microscopy has been reported on the group to date.

Cross-sectional images of eucheumatoid tissue structure and detailed descriptions of cellular arrangement are often included in taxonomic descriptions, including for *Eucheuma denticulatum* (as *E. spinosum*[105]), *Kappaphycus alvarezii*[104,105] and *Eucheumatopsis isiformis* (as *Eucheuma isiforme*[106]). Doty and Norris[107] also describe cellular arrangement in some detail in their key to commercial eucheumatoids, though include no images. These may give insight into the structure of healthy tissue; however, the study of pathology and disease diagnosis in animals and plants relies on histopathology, and observation of the disruption of normal processes. Furthering our understanding of detailed cellular arrangement and tissue structure in both healthy and diseased eucheumatoid tissue should be prioritized in order to associate microscopic changes with gross disease signs.

10 | A HOLOBIONT APPROACH TO UNDERSTANDING DISEASE PROCESSES

To untangle the processes behind the development of IID, we must also further our understanding of the diversity of taxa associated

with affected hosts, in both healthy and diseased states.[108] Like other macroorganisms, seaweeds are holobionts, complex and dynamic assemblages of taxa forming metabolic interdependencies and symbioses as a functional unit.[109] Taxa associated with seaweed holobionts can have negative or positive effects on their host[110,111] and can function as essential symbionts for normal development at various life-history stages.[112,113] Holobiont 'breakdown', in which these relationships are disrupted by an external stressor—usually environmental conditions or invasion by a pathogen—has been linked to disease in other sessile marine organisms, including corals and sponges.[114–116] Syndromic diseases are often pathobiotic systems, in which multiple symbionts, interacting with both the host and the environment, play a range of context-dependent roles in determining health status.[108] This has been shown to be particularly applicable to coral diseases,[103,117,118] which exhibit many parallels to IID. Researchers have struggled to find a single causative agent for many coral bleaching syndromes, and stress the importance of understanding the composition of, and changes in, the coral holobiont as a whole, where multiple taxa may play a role in the development of disease signs.[117,119] Holobiont units with low levels of functional diversity (i.e. highly specialized metabolic interdependencies, the disruption of which cannot be compensated for by other taxa) are particularly susceptible to break down under conditions which induce stress in either the host or other constituent taxa.[114,120]

Nothing is known about either the taxonomic or functional structure of eucheumatoid holobionts, in either healthy individuals or those showing disease signs. Understanding the composition and functional capacities of the eucheumatoid holobiont, and how these change over cultivation cycles and in association with disease signs, are key to understanding which processes are disrupted, and the identity and function of key taxa involved.

11 | DIVERSITY AND FUNCTION OF SEAWEED HOLOBIONTS

Focus on microbial diversity in marine holobionts is largely biased towards prokaryotes, with viruses and microeukaryotes remaining understudied.[120] Microbial eukaryotes have long been known to have important interactions with red algae, particularly widely cultivated species such as *Porphyra* sensu lato, where they may function as pathogens (e.g. oomycetes *Pythium porphyrae* and *Olpidiopsis pyropiae*[121,122]) or pests, such as biofouling diatoms.[88]

Associations between endophytic fungi and macroalgae are also well known and have been observed in red, green and brown macroalgal hosts from marine environments across the world.[123] These relationships can be pathogenic, saprobic or 'asymptomatic'. Pathogenic relationships are the best studied, particularly the formation of galls, abnormal hyperplastic growth in the host tissue of brown algae in response to the presence of the fungi.[124,125] In comparison, the diversity and nature of non-pathogenic associations (which produce no external signs) are understudied.[126] A number of fungi have been linked to disease signs in eucheumatoids. Solis et al.[127] isolated three species of Ascomycota (identified as *Aspergillus* spp. and *Phoma* sp. by 18S rRNA sequencing) from IID-affected *Kappaphycus alvarezii* tissue and demonstrated the ability of these isolates to induce IID signs in healthy, non-axenic *K. alvarezii*. Dewey et al.[41] isolated *Scopulariopsis brevicaulis* and *Verticillium theobromae* from *Eucheuma denticulatum* tissue showing signs of IID. The authors demonstrated *S. brevicaulis* was able to grow on heat-treated extracts of a number of red algae, including the carrageenophyte *Chondrus crispus* from the United Kingdom. However, the ability of these isolates to cause IID-like signs in eucheumatoid tissues was not tested.

Viruses are increasingly recognized as pathogens of algae.[128,129] Lachnit et al.[130] show that viruses may indeed be an important but neglected part of the red algal holobiont, characterizing a number of viruses associated with *Delisea pulchra*, including potential pathogens. Green spot disease, affecting a number of valuable *Pyropia* crop species in Korea, has been shown to be caused by a virus, PyroV1,[131] and viruses have also been detected associated with bleaching in kelp *Ecklonia radiata*.[132] To date, there are no reports of attempts to investigate the diversity of viruses associated with eucheumatoids, either in healthy tissue or in tissue showing disease signs.

Filamentous endophytic algae are also known pathogens of a number of seaweeds. The endophytic brown alga *Streblonema aecidioides* causes thallus deformations in *Saccharina latissima*,[133] and *Blastophysa rhizopus* is a pathogenic green endophyte which leads to 'green spot rotting' in *Neodilsea yendoana*.[134] In the Asian carrageenophyte species *Chondrus ocellatus*, infections with green (*Ulvella ramosa*) and brown (*Mikrosyphar zosterae)* endophytes can co-occur and inhibit host growth.[135] Though infestation with epiphytic algae is sometimes correlated with the appearance of IID signs, the role of endophytic algae in reducing host fitness in eucheumatoids remains unexplored.

Future studies on the eucheumatoid holobiont should aim to not only explore the diversity of understudied holobiont components but also elucidate their interactions with each other. Some plant and algal parasite groups, such as phytomyxids, are known to function as vectors for a range of plant pathogenic viruses (e.g. the vectoring of beet soil-borne virus and beet necrotic yellow vein virus to sugar beet *Beta vulgaris* by the phytomyxid *Polymyxa betae*[136]), and interactions such as these remain largely unexplored in algal hosts. Similarly, hyperparasitism, in which a parasite is itself host to other parasitic taxa, is largely overlooked in plant and macroalgal holobionts.[108]

Interactions between seaweeds and their associated microbiota are increasingly investigated using methodologies seeking to determine the action of the metabolites produced. Epiphytic prokaryote communities associated with macroalgae are known to be highly specific, and their composition is driven by function rather than taxonomy in response to the physiological and biochemical properties of the host.[137] As such, bacterial community composition will vary across seasons, life-history stage, and different parts of the thallus.[110,138] Seaweed 'epimicrobiomes' regulate interactions between their host algae and other taxa in the water column and can have either protective or pathogenic effects.[110,111] An important role of these taxa is the production of antifouling compounds

inhibiting further colonization of host algal thalli. These compounds can either be targeted (e.g. anti-fungal, anti-bacterial) or general, for example the production of oxidative bursts to defend against opportunistic pathogens.[138,139] Conversely, seaweed epimicrobiomes have also been shown to selectively recruit beneficial microbes which may confer protection against the development of disease signs.[110] Disruption or loss of the normal functioning of these epimicrobial communities may leave hosts more susceptible to colonization by epiphytic algae or opportunistic pathogens. Endophytic fungi are also known to produce a number of anti-microbial, anti-fungal and repellent metabolites, which deter the further colonization of the host thalli, thereby reducing competition and tissue damage caused by grazing metazoans.[140] The production of these compounds may be influenced by environmental conditions.[126]

12 | 'DOMESTICATION' AND HOST GENETICS

The host is an important part of any holobiont system, and as such, cultivated eucheumatoids may be considered as 'domesticated' holobionts. Though originally hugely productive, imported strains from the Philippines cultivated in Zanzibar have shown a significant decline in productivity after several decades of use, potentially suggesting an inability to adapt in the long term to differing environmental conditions (particularly increased water temperature,[18] or a decline in host genetic diversity as a result of repeated clonal propagation.[141–143] The only genetic data available for eucheumatoids are genotype barcoding regions (particularly short mitochondrial markers and plastid *rbcL*[144–146]) and a very small number of plastid and mitochondrial genomes for a limited number of individuals,[147–149] so it is unclear whether this 'loss of vigour' can be attributed to reduced host genetic diversity, poor adaptation to cultivation conditions, changes in holobiont structure or a combination of these factors.

The effect of propagation method on holobiont structure in commercially exploited seaweeds has not been investigated. There is evidence that at least a portion of coral symbionts are vertically transmitted, with others horizontally transmitted from the water column,[150] and studies in agriculture crops suggest fewer symbionts are inherited using micropropagation methods. It remains unclear how clonally propagated individuals acquire core symbiont taxa.[151]

Sampling campaigns aiming to further our understanding of the genetic diversity of eucheumatoids frequently include surveys of wild specimens and uncultivated species in Southeast Asia (e.g. references [144,145]). However, all reports of IID in the literature come from seaweed farms, unmanaged populations of introduced cultivars on abandoned farms[84] or laboratory cultivation experiments (Table 1). It is not clear whether this apparent resistance to IID in wild individuals and uncultivated species suggests a loss of resilience in cultivated strains (either in the host or associated microbiota), or whether the increased exposure to environmental stressors in cultivation settings (higher water temperatures, low light levels and reduced water movement) influences IID incidence. It should also be noted that the eucheumatoid cultivars used in aquaculture are often cultivated in locations geographically distant from their natural native range (many cultivars originate from wild stocks collected in the Philippines[18,19]). This is particularly true of cultivars imported into African countries, where they may be poorly adapted to local environmental conditions. Similarly, experimental attempts at the cultivation of *Kappaphycus alvarezii* in the Caribbean and South America (including Ecuador, Mexico, Panama, Venezuela) using seedlings originating from the Philippines have proven to be largely unsuccessful due to development of IID.[152]

Halling et al.[146] and Eggertson and Halling[153] recognize the need to identify eucheumatoid strains native to East Africa which may be suitable for use in aquaculture to reduce reliance on imported cultivars. Unlike many widely used cultivars,[142] these strains may be capable of sexual reproduction, and so may be able to avoid many of the problems associated with the 'loss of vigour' observed in clonally propagated individuals.

13 | MODERN METHODOLOGICAL APPROACHES TO SYNDROMIC DISEASE

Microbial diversity associated with IID has only been studied using culture-dependent methods, with the exception of a recent study utilizing Illumina 16S amplicon sequencing to compare prokaryote communities associated with a very small number of healthy and IID-affected eucheumatoids and planktonic communities,[79] and a further study by Riyaz et al.[44] which investigated prokaryote communities associated with diseased *Kappaphycus alvarezii* tissues only. A number of purported 'pathogens' apparently able to induce disease in healthy tissue have been identified.[37,76] Largo et al.[76] highlighted a major limitation of these culture-dependent methodologies, demonstrating that choice of culture medium influences the number and diversity of bacterial colonies obtained from affected tissues. As a result, microbial diversity is almost certainly underestimated, as demonstrated by the very small number of taxa isolated in IID studies to date. Furthermore, only a very small proportion of bacteria are readily culturable *in vitro*,[154] with many more having specific requirements limiting their culturability, or specialist life stages such as viable but non-culturable (VBNC) states, in which they are metabolically active but cannot be cultured. VBNC states have been observed in a number of disease-associated *Vibrio* and *Pseudomonas* strains,[155–157] and so culture-dependent methodologies attempting to compare microbiota between healthy and diseased tissue may misleadingly propose that these strains are not present as part of the normal microbiota. In holobiont systems with high levels of metabolic interdependency, obligate symbionts may be difficult to culture without knowledge of their required metabolites.[158]

The advent of more accessible and affordable high-throughput sequencing platforms (e.g. 454 and Illumina) has led to increased utilization of 16S rRNA barcoding regions to characterize microbial communities. Culture-independent 16S-targeted amplicon studies

consistently identify high levels of prokaryote diversity associated with marine holobionts[118,159,160] and are increasingly applied to seaweeds.[161,162] 16S metabarcoding has demonstrated shifts in the structure of prokaryote communities and their predicted function associated with mechanical damage to some *Gracilaria* species caused by grazers,[163] and pathogen-mediated bleaching of *Delisea pulchra*.[102]

However, 16S amplicon sequencing also has a number of biases and limitations. DNA extraction and amplification biases may mean that not all prokaryote taxa present, particularly those forming more resistant cell walls, or genetically divergent taxa including archaea, are accurately represented in studies using standard extraction methods and general 'prokaryote' primers.[159,164,165] Metabarcoding methods also rely on well populated and annotated reference databases for taxonomic assignation and prediction of functional capacity.[166,167]

Metagenomic (shotgun) sequencing methods do not selectively target barcoding regions and therefore are not subject to the same primer biases as amplicon-based methodologies. Instead, total genomic DNA is sequenced, allowing for greater insight into the diversity of taxa associated with hosts, including prokaryotes, eukaryotes and viruses. In comparison, metabarcoding methodologies are limited to the target taxa, which in most cases so far has been prokaryotes. Direct pan-genomic sequencing using metagenomics allows for more comprehensive study of the functional capacity of holobionts when compared to 16S metabarcoding, which relies on predictions based on the taxonomic affiliation of abundant partial 16S sequence types to assign function (e.g. PICRUSt2[167]; Tax4Fun[168]). It should be remembered, however, that metagenomic analysis pipelines also depend on well annotated and taxonomically unbiased reference databases and requires much deeper sequencing of each sample than for metabarcoding, while retaining sufficient numbers of samples to enable robust statistical interpretation of the results.[169]

Metagenomics is a powerful tool for tracking changes in the structure and functional capacity of stressed holobionts, as demonstrated by Fan et al.[114], who linked changes in the taxonomic composition and functional capacity of the holobiont of the sponge *Rhopaloides odorabile* to the development of disease signs and subsequent breakdown of symbioses. They also demonstrate that the dominant bacterial taxa in diseased tissues did not possess identifiable virulence mechanisms (and so are likely not true pathogens), instead appearing to be fast-growing nutrient scavengers including *Vibrio* and *Pseudoalteromonas*, bacterial groups which are often associated with IID in eucheumatoids.

The application of a combination of molecular methods ('multi-omics') has already offered considerable insight into holobiont interactions in another red algal bleaching disease, the pathogen-mediated bleaching of *Delisea pulchra*. 16S sequencing[102] and metagenomics[170] have elucidated shifts in prokaryote community structure and functional capacity associated with bleaching, and transcriptomics[171] has offered some insight into host-pathogen interactions in this disease system. Metagenomics has also been used to examine how bacteria within the *D. pulchra* holobiont may interact with each other, and the effect this has on the development of bleaching signs.[172] However, histology and other microscopy techniques remain underutilized tools in the study of seaweed diseases, particularly multi-omic studies. Pairing high-quality microscopy with selected high-throughput sequencing and metabolomic techniques will allow for a greater understanding of the disease processes behind observed gross disease signs and allow for detailed characterization of both healthy and diseased states.

14 | DEVELOPING A TOOLKIT FOR THE STUDY OF SYNDROMIC DISEASES IN SEAWEED

Seaweeds lag behind other sessile marine holobionts in terms of our understanding of their structure, function and disease processes. To date, there has been very limited application of complementary methodologies (i.e. molecular and microscopic techniques), an approach which has been used to great effect to elucidate the contributing factors to coral bleaching syndromes.

Histological techniques are frequently utilized in coral disease research, allowing the visualization of gross lesions and tissue loss using appropriate stains,[173,174] with more recent studies combining them with high-throughput 16S amplicon sequencing[175]. Bourne et al.[176] highlight the importance of this kind of multidisciplinary approach–combining histological and molecular tools in order to assess interactions between host, pathogenic agents and the environment–particularly when examination of gross signs can lead to inconsistent or misleading in-field diagnosis. Greene et al.[177] propose a toolkit for the standard sampling and processing of corals, aimed at facilitating the quantitative characterization of coral health and improved ability to describe and model impacts on coral reef systems. The development of a similar set of complementary protocols aiming to characterize health and disease in cultivated eucheumatoids will allow for the more robust comparison between healthy and diseased holobionts of cultivated strains, and a greater understanding of the link between shifts in holobiont structure and function observed microscopic processes in host tissues and environmental parameters. Using the terminology of Bass et al.[108], identifying whether specific pathogens, a defined 'pathobiome', or a range of pathobiomes are associated with the disease defined as IID will be necessary before a definitive case definition can be constructed for this disease state.

15 | CONCLUSION AND RESEARCH PRIORITIES

The biggest challenge to our understanding of IID is elucidating the disease processes resulting in the observed signs; essentially, a unified case description of IID is not available. Future research must urgently address the lack of microscopic evidence associated with IID and be accompanied by a greater effort to understand the normal

structures of healthy tissues in order to correlate microscopic observations with gross signs. The application of complementary methods, which should include high-quality microscopy and appropriate molecular methods such as high-throughput sequencing (metagenomics and/or 16S metabarcoding, as well as more directly functionally informative methods such as RNASeq), to as broad a range of eucheumatoid species, cultivars and farming techniques as possible, should be used to link disease signs to potential causative factors, such as holobiont breakdown or invasion by a specific pathogen, and facilitate investigation of the mechanism by which these factors act. Once this evidence is available, it will be possible to determine whether a unified case description for IID as the term is currently applied is possible, or whether the condition encompasses a range of microscopically diverse bleaching syndromes, potentially with a number of different causes, which may explain the inconsistency in observed signs noted in Table 1.

Identification of IID causative factor(s) and their interaction with associated environmental stressors will in turn allow for the development of evidence-based guidelines for seedling propagation, farm management and disease risk monitoring, aiming to retain host and symbiont genetic and functional diversity and optimize cultivation techniques to limit exposure to stressors. Taken together, these measures will be necessary to support the sustainable production of eucheumatoid seaweeds in the global aquaculture industry.

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REFERENCES

1. Stentiford GD, Sritunyalucksana K, Flegel TW, et al. (2017) New paradigms to help solve the global aquaculture disease crisis. *PLoS Pathog*. 2017;13:e1006160.
2. Ward GM, Faisan JP Jr, Cottier-Cook EJ, et al. A review of reported seaweed diseases and pests in aquaculture in Asia. *J World Aquacult Soc*. 2020;51(4):815-828.
3. Cottier-Cook EJ, Nagabhatala N, Badis Y, Campbell M, Chopin T, Dai W, et al. Safeguarding the future of the global seaweed aquaculture industry. United Nations University (INWEH) and Scottish Association for Marine Science Policy Brief. 2016; ISBN 978-92-808-6080-1.
4. Campbell I, Kambey C, Mateo J, et al. Biosecurity policy and legislation for the global seaweed aquaculture industry. *J Appl Phycol*. 2020;32(4):2133-2146.
5. FAO. *The State of World Fisheries and Aquaculture 2020*. Sustainability in Action. Rome: FAO; 2020. https://doi.org/10.4060/ca9229en
6. Kambey CSB, Campbell I, Sondak CFA, Nor ARM, Lim PE, Cottier-Cook EJ. An analysis of the current status and future of biosecurity frameworks for the Indonesian seaweed industry. *J Appl Phycol*. 2020;32(4):2147-2160.
7. Rusekwa SB, Campbell I, Msuya ME, Buriyo AS, Cottier-Cook EJ. Biosecurity policy and legislation of the seaweed aquaculture industry in Tanzania. *J Appl Phycol*. 2020;32(6):4411-4422.
8. Kelly ELA, Cannon AL, Smith JE. Environmental impacts and implications of tropical carrageenophyte seaweed farming. *Conservation Biol*. 2020;34(2):326-337.
9. de la Torre-Castro M, Rönnbäck P. Links between humans and seagrasses – an example from tropical East Africa. *Ocean Coast Manag*. 2004;47(7–8):361-387.
10. Barrios J, Bolaños J, López R. Blanqueamiento de arrecifes coralinos por la invasion de *Kappaphycus alvarezii* (Rhodophyta) en isla cubagua, Estado Nueva Esparta, Venezuela (Spanish language). *Biol Inst Oceanogr Venezuela*. 2007;46(2):147-152.
11. Chandrasekaran S, Nagendran NA, Pandiaraja D, Krishnankutty N, Kamalakannan B. Bioinvasion of *Kappaphycus alvarezii* on corals in the Gulf of Mannar, India. *Curr Sci*. 2008;94(9):1167-1172.
12. Sellers AJ, Saltonstall K, Davidson TM. The introduced alga *Kappaphycus alvarezii* (Doty ex P.C. Silva, 1996) in abandoned cultivation sites in Bocas del Toro, Panama. *Bioinvasions Rec*. 2015;4(1):1-7.
13. Conklin EJ, Smith JE. Abundance and spread of the invasive red algae, *Kappaphycus* spp., in Kane'ohe Bay, Hawai'i and an experimental assessment of management options. *Biol Invasions*. 2005;7(6):1029-1039.
14. Stentiford GD, Bateman IJ, Hinchcliffe SJ, et al. Sustainable aquaculture through the One Health lens. *Nat Food*. 2020;1:468-474.
15. Trono GC. *Eucheuma Farming in the Philippines*. University of the Philippines Natural Science Research Center; 1974:13.
16. Doty MS, Alvarez VB. Status, problems, advances and economics of *Eucheuma* farms. *Mar Technol Soc J*. 1975;9:30-35.
17. Hurtado AQ, Neish IC, Critchley AT. Phyconomy: the extensive cultivation of seaweeds, their sustainability and economic value, with particular reference to important lessons to be learned and transferred from the practice of eucheumatoid farming. *Phycologia*. 2019;58(5):472-483.
18. Largo DB, Msuya FE, Menezes A. *Understanding the Diseases and Die-Off of Seaweeds in Zanzibar*. FAO Fisheries and Aquaculture Technical Paper. FAO. 2020;662:56 pp.
19. Hayashi L, Hurtado AQ, Msuya FE, Bleicher-Lhonneur G, Critchley AT. A review of *Kappaphycus* farming: prospects and constraints. In: Seckbach J, Einav R, Israel A, eds. *Seaweeds and their Role in Globally Changing Environments. Cellular Origin, Life in Extreme Habitats and Astrobiology* Vol 15.Springer; 2010.
20. Rantetondok A, Latama G. Epiphytic and ice-ice diseases of seaweed, *Kappaphycus alvarezii* and its effect on growth rate and carrageenan quality, *International. J Aquacult*. 2017;7(21):134-138.
21. Mendoza WG, Montaño NE, Ganzon-Fortes ET, Villanueva RD. Chemical and gelling profile of *ice-ice* infected carrageenan from *Kappaphycus striatum* (Schmitz) Doty "*sacol*" strain (Solieriaceae, Gigartinales, Rhodophyta). *J Appl Phycol*. 2003;14(5):409-418.
22. Loureiro RR, Reis RP, Critchley AT. In vitro cultivation of three *Kappaphycus alvarezii* (Rhodophyta, Areschougiaceae) variants (green, red and brown) exposed to a commercial extract of the brown alga *Ascophyllum nodosum* (Fucaceae, Ochrophyta). *J Appl Phycol*. 2009;22(1):101-104.
23. Amiluddin NM. *Study of Growth of Carrageenan Content of SEAWEED* Kappaphycus alvarezii *Affected by Ice-Ice Disease in the Waters of Puri Island, Thousand Islands (Indonesian language)*. Thesis for MS. Graduate School of the Institute of Agriculture; 2007.
24. Vairappan CS, Chung CS, Matsunaga S. Effect of epiphyte infection on physical and chemical properties of carrageenan produced by *Kappaphycus alvarezii* Doty (Solericeae, Gigartinales, Rhodophyta). *J Appl Phycol*. 2014;26(2):923-931.
25. Samonte GPB. Economics of *Kappaphycus* spp. seaweed farming with special reference to the Central Philippines. In: Hurtado

AQ, Critchley AT, Neish IC, eds. *Tropical Seaweed Farming: Trends, Problems and Opportunities*. Springer; 2017:147-154.
26. Lunkapis GJ, Danny HW. Preliminary impact assessment of seaweed cultivation by the coastal communities in Sabah, Malaysia. *Trans Sci Technol*. 2016;3:413-420.
27. Msuya FE, Hurtado AG. The role of women in seaweed aquaculture in the Western Indian Ocean and South-East Asia. *Eur J Phycol*. 2017;52(4):482-494.
28. Yulianto K. 2004) Fenomena faktor pengontrol penyebab kerugian pada budidaya karaginofit di Indonesia (Indonesian language. *Oseana*. 2004;29(2):17-23.
29. Vairappan CS, Chung CS, Hurtado AQ, Soya FE, Bleicher-Lhonneur G, Critchley A. Distribution and symptoms of epiphyte infection in major carrageenophyte-producing farms. *J Appl Phycol*. 2008;20(5):477-483.
30. Laisow S. *Dampak budidaya rumput laut terhadap tingkat pendpatan masyarakat Kecamatan Dullah Utara, Kota Tual (Indonesian language)*. Master's Thesis. Universitas Terbuka; 2013.
31. Galgo CR, Abamo AP. Profitability analysis of seaweed (*Eucheuma denticulatum* (NL Burman) FS Collins & Hervey) enterprise in Dawahon Island Bato, Leyte: a green value chain approach. *JST*. 2014;4(1):63-74.
32. Eranza DRD, Bahron A, Alin J. Sustaining seaweed farming in Malaysia. *Int J Bus Manag*. 2015;3(3):201.
33. Eranza DRD, Bahron A, Alin J, Mahmud R, Malusirang SR. Ongoing assessment of issues in the seaweed farming industry in Sabah, Malaysia. *JAAAB*. 2017;4:49-60.
34. Trono GC, Valdestamon RG. New aspects in the ecology and culture of *Kappaphycus* and *Eucheuma*. *Kor J Phycol*. 1994;9(2):205-216.
35. Aris M. *Identification, Bacterial Pathogenicity Dan the Application of 16S-rRNA Gene for the Detection of Ice Ice Disease in the Cultivation of Seaweed (*Kappaphycus alvarezii*) (Indonesian language)*. PhD thesis, Graduate School of the Institute of Agriculture; 2011.
36. Saraswati SA. Darmasetiyawana IMS (2016) Identifikasi Bakteri pada Rumput Laut *Eucheuma spinosum* yang terserang penyakit *Ice-ice* di Perairan Pantai Kutuh (Indonesian language). *J Mar Aquat Sci*. 2016;2:11-15.
37. Azizi A, Hanafi NM, Basiran MN, Teo CH. Evaluation of disease resistance and tolerance to elevated temperature stress of the selected tissue-cultured *Kappaphycus alvarezii* Doty 1985 under optimised laboratory conditions. *Biotech*. 2018;8(8):321.
38. Mariño M, Breckwoldt A, Teichberg M, Kase A, Reuter H. Livelihood aspects of seaweed farming in Rote Island, Indonesia. *Mar Policy*. 2019;107:103600.
39. Riyaz SUM, Prabhu NS, Nalini S, Tejaswini G, Christy HJ, Inbakandan D. Microbiome identification from ice-ice disease infected *Kappaphycus alvarezii* using 16S rRNA gene sequencing analysis and in silico docking studies against carrageenan degrading bacteria. *Biocatal Agric Biotechnol*. 2020;27:101707.
40. Uyenco FR, Saniel LS, Jacinto GS. The "Ice-ice" problem in seaweed farming. In: *Proceedings of the 10th International Seaweed Symposium*. Walter de Gruyter; 1981:625-630.
41. Dewey FM, Hunter-Blair CM, Banbury GH, Donnelly KA. Isolation of *Scopulariopsis brevicaulis* from *Eucheuma striatum* and its ability to degrade seaweeds and their soluble products. *Trans Br Mycol Soc*. 1984;83(4):621-629.
42. Irmawati Y, Sudjiro F. Infection *Vibrio* sp. bacteria on *Kappaphycus* seaweed varieties brown and green. *IOP Conf. Ser.: Earth Environ. Sci*. 2017;89:0126016.
43. Aris M, Labuena R. Horizontal transmission of ice-ice disease in seaweed *Kappaphycus alvarezii* (Doty) L. M. Liao (Gigartinales, Rhodophyta). *Hidrobiológica* 2020;30(2):155-161.
44. Riyaz SUM, Inbakandan D, Manikandan D, et al. Microbiome in the *ice-ice* disease of the farmed red algae *Kappaphycus alvarezii* and degradation of extracted food carrageenan. *Food Biosci*. 2021;42:101138.
45. Maulani RK, Achmad M, Atama G. Karakteristik jaringan secara histologi dari strain rumput laut (*Kappaphycus alvarezii*) yang terinfeski penyakit ice-ice (Indonesian language). *Torani J Fish Mar Sci*. 2017;1(1). https://doi.org/10.35911/torani.v1i1.3796.
46. Sade A, Ali I, Ariff MRM. The seaweed industry in Sabah, East Malaysia. *Jati*. 2006;11:97-107.
47. Pang T, Liu J, Liu Q, Li H, Li J. Observations on pests and diseases affecting a eucheumatoid farm in China. *J Appl Phycol*. 2015;27(5):1975-1984.
48. Tisera WL, Naguit MRA. *Ice-ice* disease occurrence in seaweed farms in Bais Bay, Negros Oriental and Zamboanga Del Norte. *The Threshold*. 2009;(4):1-16. (not peer reviewed).
49. Perez-Enriquez R, Growth of *Eucheuma isiforme* (C. Agardh) J. Agardh on experimental rafts off the coast of Yucatan State, Mexico. *J Appl Phycol*. 1996;8(1):27-28.
50. Largo DB, Fukami K, Adachi M, Nishijima T. Direct enumeration of bacteria from macroalgae by epifluorescence microscopy as applied to the fleshy red algae *Kappaphycus alvarezii* and *Gracilaria* spp. (Rhodophyta). *J Appl Phycol*. 1997;14(5):409-418.
51. Nasmia RS, Rantetondok A, Zainuddin EN. Characterisation and identification of bacteria isolated from seaweed *Gracilaria verrucosa* (Linn., 1758) infected by ice-ice. *Int J Aquacult*. 2014;4(23):1-7.
52. Zainuddin EN, Anshary H, Huyyirnah H, Hiola R, Baxa DV. Antibacterial activity of *Caulerpa racemosa* against pathogenic bacteria promoting "ice-ice" disease in the red alga *Gracilaria verrucosa*. *J Appl Phycol*. 2019;31(5):3201-3212.
53. Waters TJ, Lionata H, Prasetyo Wibowo T, et al. *Coastal Conservation and Sustainable Livelihoods through Seaweed Aquaculture in Indonesia: A Guide for Buyers, Conservation Practitioners, and Farmers. Version 1*. The Nature Conservancy. 2019 (not peer reviewed).
54. Largo DB, Fukami K, Nishijima T, Ohno M. Laboratory-induced development of the *ice-ice* disease of the farmed red algae *Kappaphycus alvarezii* and *Eucheuma denticulatum* (Solieriaceae, Gigartinales, Rhodophyta). *J Appl Phycol*. 1995;7(6):539-543.
55. Alibon RD, Gonzales JMP, Ordoyo AET, Echem RT. Incidence of ice-ice disease associated with *Kappaphycus alvarezii* in the seaweed farms in Zamboanga Peninsula, Mindanao, Philippines. *SSR Inst Int J Life Sci*. 2019;5(1):2148-2155.
56. Maryunus RP. Pengendalian penyakit ice-ice budidaya rumput laur, *Kappaphycus alvarezii:* korelasi musim dan manipulasi terbatas lingkungan (Indonesian language). *Jurnal kebijakan perikanan Indonesia*. 2018;10(1):1-10.
57. Largo DB, Fukami K, Nishijima T. Time-dependent attachment mechanism of bacterial pathogen during ice-ice infection in *Kappaphycus alvarezii* (Gigartinales, Rhodophyta). *J Appl Phycol*. 1999;11(1):129-136.
58. Mollion J, Braud J-P. A *Eucheuma* (Solieriaceae, Rhodophyta) cultivation test on the south-west coast of Madagascar. *Hydrobiologia*. 1993;260/261:373-378.
59. de San M. *The Farming of Seaweeds*. SmartFish Programme Report. 2012;SF/2012/28.
60. Wakibia JG, Bolton JJ, Keats DW, Raitt LM. Factors influencing the growth rates of three commercial eucheumoids at coastal sites in southern Kenya. *J Appl Phycol*. 2006;18(3–5):565-573.
61. Kim JK, Yarish C, Hwang EK, Park M, Kim Y. Seaweed aquaculture: cultivation technologies, challenges and its ecosystem services. *Algae*. 2017;32(1):1-13.
62. Luhan MRJ, Avañcena SS, Mateo JP. Effect of short-term immersion of *Kappaphycus alvarezii* (Doty) Doty in high nitrogen on the growth, nitrogen assimilation, carrageenan quality, and occurrence of "ice-ice" disease. *J Appl Phycol*. 2015;27(2):917-922.
63. Agustina SS, Marlan WF. Identifikasi tingkat serangan bakteri yang menginfeksi komoditi rumput laut di perairan Teluk Tolo dan Teluk Tomini kabupaten banggai Sulawesi tengah (Indonesian language). *Jurnal Balik Diwa*. 2014;5(2):34-39.

64. Vairappan CS. Seasonal occurrences of epiphytic algae on the commercially cultivated red alga *Kappaphycus alvarezii* (Solieriaceae, Gigartinales, Rhodophyta). *J Appl Phycol*. 2006;18(3):611-617.
65. Araújo PG, Ribeiro ALNL, Yokoya NS, Fujui MT. Temperature and salinity responses of drifting specimens of *Kappaphycus alvarezii* (Gigartinales, Rhodophyta) farmed on the Brazilian tropical coast. *J Appl Phycol*. 2014;26(5):1979-1988.
66. Faisan JP Jr, Luhan MRJ, Sibonga RC, et al. Preliminary survey of pests and diseases of eucheumatoid seaweed farms in the Philippines. *J Appl Phycol*. 2021;33(4):2391-2405. https://doi.org/10.1007/s10811-021-02481-5
67. Largo DB. Recent developments in seaweed diseases. In: Hurtado AG, Guanzon NG Jr, de Castro-Mallare TR Luhan MRJ eds. *Proceedings of the National Seaweed Planning Workshop held on August 2–3, 2001, SEADFEC Aquaculture Department, Tigbauan, Iloilo*. (pp. 35-42). SEAFDEC Aquaculture Department. 2002.
68. Leonardi PI, Miravalles AB, Faugeron S, Flores V, Bentrán J, Correa JA. Diversity, phenomenology and epidemiology of epiphytism in farmed *Gracilaria chilensis* (Rhodophyta) in northern Chile. *Eur J Phycol*. 2009;41(2):247-257.
69. Mtolera MSP, Collén J, Pedersén M, Semesi AK. Destructive hydrogen peroxide production in *Eucheuma denticulatum* (Rhodophyta) during stress caused by elevated pH, high light intensities and competition with other species. *Eur J Phycol*. 1995;30(4):289-297.
70. Mtolera MSP, Collén J, Pedersén M, Ekdahl A, Abrahamsson K, Semesi AK. Stress-induced production of volatile halogenated organic compounds in *Eucheuma denticulatum* (Rhodophyta) caused by elevated pH and high light intensities. *Eur J Phycol*. 1996;31(1):89-95.
71. Ask EI, Azanza RV. Advances in cultivation technology of commercial eucheumatoid species: a review with suggestions for future research. *Aquaculture*. 2002;206(3–4):257-277.
72. Mtolera MSP. Effect of seagrass cover and mineral content on *Kappaphycus* and *Eucheuma* productivity in Zanzibar. *Western Indian Ocean J Mar Sci*. 2003;2(2):163-170.
73. Loureiro RR, Hurtado AQ, Critchley AT. Impacts of AMPEP on Epiphytes and Diseases in Kappaphycus and Eucheuma Cultivation. In: Hurtado A, Critchley A, Neish I, eds. *Tropical Seaweed Farming Trends, Problems and Opportunities. Developments in Applied Phycology*, vol 9. Springer; 2017:111-119.
74. Riyaz SUM, Nalini S, Kavitha G, Sutha SAD, Inbakandan D. Characterisation and identification of isolated bacteria from ice-ice disease infected seaweed *Kappaphycus alvarezii*. *Indian J Mar Sci*. 2019;48(8):1286-1290.
75. Butardo VM, Ganzon-Fortes ET, Silvestre VA, Bacano-Maningas MBB, Montaño MNE, Lluisma AO. Isolation and classification of culturable bacteria associated with *ice-ice* disease in *Kappaphycus* and *Eucheuma* (Rhodophyta, Solieriaceae). *Philipp Scient*. 2003;40:223-237.
76. Largo DB, Fukami K, Nishijima T. Occasional pathogenic bacteria promoting *ice-ice* disease in the carrageenan-producing red algae *Kappaphycus alvarezii* and *Eucheuma denticulatum* (Solieriaceae, Gigartinales, Rhodophyta). *J Appl Phycol*. 1995;7(6):545-554.
77. Thompson FL, Gevers D, Thompson CC, et al. Phylogeny and molecular identification of Vibrios on the basis of multilocus sequence analysis. *Appl Env Microbiol*. 2005;71(9):5107-5115.
78. Syafitri E, Prayitno SB, Ma'ruf WF, Radjasa OK. Genetic diversity of the causative agent of ice-ice disease of the seaweed *Kappaphycus alvarezii* from Karimunjawa Island, Indonesia. *IOP Conf Ser: Earth Environ Sci*. 2016;55:012044.
79. Kopprio GA, Cuong LH, Luyen ND, et al. Carrageenophyte-attached and planktonic bacterial communities in two distinct bays of Vietnam: eutrophication indicators and insights on ice-ice disease. *Ecol Indic*. 2021;121:107067.
80. Wulandari SA, Isdiantoni PEN. Analisis fisika-kimia perairan dan komunitas bakteri terkait kemunculan penyakit ice-ice pada rumput laut (*Kappaphycus alvarezii*). (Indonesian language). *SEAJ*. 2020;2(2):66-78.
81. Atewberhan M, Rougier A, Rakotomahazo C. Influence of environmental factors and farming technique on growth and health of farmed *Kappaphycus alvarezii* (cottonii) in south-west Madagascar. *J Appl Phycol*. 2015;27(2):923-934.
82. Aris M, Ibrahim TA. Laju transmisi penyakit ice-ice pada rumput laut *Kappaphycus alvarezii* berdasarkan jarak tanam dengan metode longline (Indonesian language). *Budidaya Perairan*. 2020;8(2):82-90.
83. Msuya FE. Effects of stocking density and additional nutrients on growth of the commercially farmed seaweeds *Eucheuma denticulatum* and *Kappaphycus alvarezii* in Zanzibar Tanzania. *TaJONAS*. 2013;4(1):605-612.
84. Msuya FE, Porter M. Impact of environmental changes on farmed seaweed and farmers: the case of Songo Songo Island, Tanzania. *J Appl Phycol*. 2014;26(5):2135-2141.
85. Nor ARM. *The Effect of Seaweed Aquaculture Cooperatives on Sustainable Livelihoods in Malaysia*. PhD Thesis. :Newcastle University; 2018.
86. Arisandi A, Farid A. Dampak factor ekologis terhadap sebaran penyakit ice-ice (Indonesian language). *Jurnal Kelautan*. 2014;7:20-25.
87. Kambey CSB, Campbell I, Cottier-Cook E. Seaweed Aquaculture: a preliminary assessment of biosecurity measures for controlling disease and pest outbreaks of a Kappaphycus farm. *J Appl Phycol*. 2021. https://doi.org/10.1007/s10811-021-02530-z
88. Kim GH, Moon K-H, Kim J-Y, Shim J, Klochkova TA. A revaluation of algal diseases in Korean *Pyropia* (*Porphyra*) sea farms and their economic impact. *Algae*. 2014;29(4):249-265.
89. Arisandi A, Marsoedi NH, Sartimbul A. Velocity and infection percentage of ice-ice disease on *Kappaphycus alvarezii* in Bluto Beach Sumenep (Indonesian language). *JIPK*. 2011;3:47-51.
90. Calvo F, Karras BT, Phillips R, Kimball AM, Wilf F. Diagnoses, syndromes, and diseases: a knowledge representation problem. In: *American Medical Informatics Association Annual Symposium Proceedings*; 2003:802.
91. Mera H, Bourne DG. Disentangling causation: complex roles of coral-associated microorganisms in disease. *Environ Microbiol*. 2018;20(2):431-449.
92. Meyer JL, Castellanos-Gell J, Aeby GS, Häse CC, Ushijima B, Paul VJ. Microbial community shifts associated with the ongoing stony coral tissue loss disease outbreak on the Florida Reef Tract. *Front Microbiol*. 2019;10:2244.
93. Aronson RB, Precht WF. White-band disease and the changing face of Caribbean coral reefs. In: Porter JW, ed. *The Ecology and Etiology of Newly Emerging Marine Diseases*. Netherlands; 2001:25-38.
94. Great Barrier Reef Marine Park Authority. *Final Report: 2016 Coral Bleaching Event on the Great Barrier Reef*. GBRMPA; 2017. ISBN 9780995373167.
95. Hou D, Huang Z, Zeng S, et al. Intestinal bacterial signatures of white faeces syndrome in shrimp. *Appl Microbiol Biotechnol*. 2018;102(8):3701-3709.
96. Alfiansah YR, Peters S, Harder J, Hassenrück C, Gärdes A. Structure and co-occurrence patterns of bacterial communities associated with white faeces disease outbreaks in Pacific white-leg shrimp *Penaeus vannamei* aquaculture. *Sci Rep*. 2020;10:11980.
97. Mastan SA. (2015) Incidences of white faeces syndrome (WFS) in farm-reared shrimp, *Litopenaeus vannamei*, Andhra Pradesh. *Indo Am J Pharm*. 2015;5(09):3044-3047.
98. Aranguren Caro LF, Mai HN, Pichardo O, Cruz-Flores R, Hanggono B, Dhar AK. Evidences supporting *Enterocytozoon hepatopenaei* association with white feces syndrome in farmed *Penaeus vannamei* in Venezuela and Indonesia. *Dis Aquat Org*. 2020;141(1):71-78.
99. Tangprasittipap A, Srisala J, Chouwdee S, et al. The microsporidian *Enterocytozoon hepatopenaei* is not the cause of white faeces

syndrome in whiteleg shrimp *Penaeus (Litopenaeus) vannamei*. *BMC Vet Res*. 2013;9:139.
100. Case RJ, Longford SR, Campbell AH, et al. Temperature induced bacterial virulence and bleaching disease in a chemically defended marine macroalga. *Environ Microbiol*. 2011;13(2):529-537.
101. Fernandes N, Case RJ, Longford SR, et al. Genomes and virulence factors of novel bacterial pathogens causing bleaching disease in the marine alga *Delisea pulchra*. *PLoS One*. 2011;6(12):e27387.
102. Zozaya-Valdes E, Egan S, Thomas T. A comprehensive analysis of the microbial communities of healthy and diseased marine macroalgae and the detection of known and potential bacterial pathogens. *Front Microbiol*. 2015;6:146.
103. Sutherland KP, Berry B, Park A, et al. Shifting white pox aetiologies affecting *Acropora palmata* in the Florida Keys, 1994–2014. *Philos Trans R Soc Long B*. 2016;371(1689):20150205.
104. Doty MS. *Eucheuma alvarezii* sp. nov. (Gigartinales, Rhodophyta) from Malaysia. In: Abbot IA, Norris JN, eds. *Taxonomy of Economic Seaweeds with Reference to some Pacific and Caribbean species*. Report No. T-CSGCP-011. California Sea Grant College Program. 1985;37-45.
105. Doty MS. Prodromus ad Systematica Eucheumatoideorum: A tribe of commercial seaweeds related to *Eucheuma*. In: Abbott IA, ed. *Taxonomy of economic seaweeds with reference to some Pacific and Caribbean Species*, vol. II. California Sea Grant College Program; 1988:59-208.
106. Cheney DP, The genus Eucheuma J. Agardh in Florida and the Caribbean. In: Abbott IA, ed. *Taxonomy of economic seaweeds with reference to some Pacific and Caribbean Species* Vol. II. California Sea Grant College Program; 1988: 209-219.
107. Doty MS, Norris JN. Eucheuma species (Solieriaceae, Rhodophyta) that are major sources of carrageenan. In: Abbot IA, Norris JN, senior eds. *Taxonomy of economic seaweeds with reference to some Pacific and Caribbean* Species Report No. T-CSGCP-011. California Sea Grant College Program; 1985;47-61.
108. Bass D, Stentiford GD, Wang H-C, Koskella B, Tyler C. The Pathobiome in animal and plant diseases. *Trends Ecol Evol*. 2019;34(11):996-1008.
109. van der Loos LM, Eriksson BK, Falcão SJ. The macroalgal holobiont in a changing sea. *Trends Microbiol*. 2019;27(7):635-650.
110. Egan S, Harder T, Burke C, Steinberg P, Kjelleberg S, Thomas T. The seaweed holobiont: understanding seaweed-bacteria interactions. *FEMS Microbiol Rev*. 2013;37(3):462-476.
111. Saha M, Weinberger F. Microbial, "gardening" by a seaweed holobiont: surface metabolites attract protective and deter pathogenic epibacterial settlement. *J Ecol*. 2019;107(5):2255-2265.
112. Lemay MA, Martone PT, Hind KR, Lindstrom SC, Parfrey LW. Alternative life history phases of a common seaweed have distinct microbial surface communities. *Molec. Ecol*. 2017;27(17):3555-3568.
113. Tapia JE, González B, Goulitquer S, Potin P, Correa JA. Microbiota influences morphology and reproduction of the brown alga *Ectocarpus* sp. *Front Microbiol*. 2016;7:197.
114. Fan L, Liu M, Simister R, Webster SN, Thomas T. Marine microbial symbiosis heats up: the phylogenetic and functional response of a sponge holobiont to thermal stress. *ISME J*. 2013;7(5):991-1002.
115. Ramsby BD, Hoogenboom MO, Whalan S, Webster NS. Elevated seawater temperature disrupts the microbiome of an ecologically important bioeroding sponge. *Mol Ecol*. 2018;27(8):2124-2137.
116. Sheridan C, Kramarsky-Winter E, Sweet M, Kushmaro A, Costa LM. Diseases in coral aquaculture: causes, implications and preventions. *Aquaculture*. 2013;396–399:124-134.
117. Sweet MJ, Bulling MT. On the importance of the microbiome and pathobiome in coral health and disease. *Front Mar Sci*. 2017;4:9.
118. Rubio-Portillo E, Kersting DK, Linares C, Ramos-Esplá AA, Antón J. Biogeographic differences in the microbiome and pathobiome of the coral *Cladocora caespitosa* in the Western Mediterranean Sea. *Front Microbiol*. 2018;9:22.
119. Rosenberg E, Kushmaro A, Kramarsky-Winter E, Banin E, Yossi L. The role of microorganisms in coral bleaching. *ISME J*. 2009;3(2):139-146.
120. Pita L, Rix L, Slaby BM, Franke A, Hentschel U. The sponge holobiont in a changing ocean: from microbes to ecosystems. *Microbiome*. 2018;6:46.
121. Takahashi M, Ichitani T, Sasaki M. *Pythium porphyrae* Takahashi et Sasaki sp. nov. causing red rot of marine red algae *Porphyra* spp. *Trans Mycol Soc Jpn*. 1977;18:279-285.
122. Klochkova TA, Shin YJ, Moon H-K, Motomura T, Kim GH. New species of unicellular obligate parasite, *Olpidiopsis pyropiae* sp. nov., that plagues *Pyropia* sea farms in Korea. *J Appl Phycol*. 2014;28(1):73-83.
123. Flewelling AJ, Currie J, Gray CA, Johnson JA. Endophytes from marine macroalgae: promising sources of novel natural products. *Curr Sci*. 2015;109(1):88-111.
124. Apt KE. Galls and tumor-like growths on marine macroalgae. *Dis Aquat Org*. 1988;4(1):211-217.
125. Neill K, Heesch S, Nelson W. Diseases, pathogens and parasites of *Undaria pinnatifida*. MAF Biosecurity Technical Paper No 2009/44. ISBN 1177-6412 (online). 2008. (not peer reviewed).
126. Suryanarayanan TS. Fungal Endosymbionts of Seaweeds. *Prog Mol Subcell Biol*. 2012;53:53-69.
127. Solis MJL, Draeger S, dela Cruz TEE. Marine-derived fungi from *Kappaphycus alvarezii* and *K. striatum* as potential causative agents of *ice-ice* disease in farmed seaweeds. *Bot Mar*. 2010;53(6):587-594.
128. Gachon CMM, Sime-Ngando T, Strittmatter M, Chambouvet A, Kim GH. Algal diseases: spotlight on a black box. *Trends Plant Sci*. 2010;15(11):633-640.
129. McKeown DA, Schroeder JL, Stevens K, et al. Phaeoviral infections are present in *Macrocystis*, *Ecklonia* and *Undaria* (Laminariales) and are influenced by wave exposure in Ectocarpales. *Viruses*. 2018;10(8):410.
130. Lachnit T, Thomas T, Steinberg P. Expanding our understanding of the seaweed holobiont: RNA viruses of the red alga *Delisea pulchra*. *Front Microbiol*. 2016;6:1489.
131. Kim GH, Klochkova TA, Lee DJ, Im SH. Chloroplast virus causes green-spot disease in cultivated *Pyropia* of Korea. *Algal Res*. 2016;17:293-299.
132. Beattie DT, Lachnit T, Dinsdale EA, Thomas T, Steinberg PD. Novel ssDNA viruses detected in the virome of bleached, habitat-forming kelp *Ecklonia radiata*. *Front Mar Sci*. 2018;4:441.
133. Peters AF, Schaffelke B. *Streblonema* (Ectocarpales, Phaeophyceae) infection in the kelp *Laminaria saccharina* (Laminariales, Phaeophyceae) in the western Baltic. *Hydrobiologia*. 1996;326/327:111-116.
134. Iima M, Tatewaki M. On the life history and host specificity of *Blastophysa rhizopus* (Codiales, Chaetosiphonaceae), an endophytic green alga from Mororan in laboratory cultures. *Jpn J Phycol*. 1987;35:241-250.
135. Gao X, Ogandaga CAM, Park SK, Oh JC, Choi HG. Algal endophytes of commercial *Chondrus ocellatus* (Gigartinaceae, Rhodophyta) from different wild populations in Korea. *J Appl Phycol*. 2019;32(1):697-703.
136. Schwelm A, Badstöber J, Bulman S, et al. Not in your usual Top 10: protists that infect plants and algae. *Mol Plant Pathol*. 2017;19(4):1029-1044.
137. Burke C, Steinberg P, Rusch D, Kjelleberg S, Thomas T. Bacterial community assembly based on functional genes rather than species. *Proc Natl Acad Sci USA*. 2011;108(34):14288-14293.
138. Hollants J, Leliaert F, de Clerck O, Willems A. What we can learn from sushi: a review on seaweed-bacterial associations. *FEMS Microbiol Ecol*. 2012;83(1):1-16.

139. Singh RP, Reddy CRK. Seaweed-microbial interactions: key functions of seaweed-associated bacteria. *FEMS Microbiol Ecol*. 2013;88(2):213-230.
140. Suryanarayanan TS, Venkatachalam A, Thirunavukkarasu N, Ravishankar JP, Doble M, Geetha V. Internal mycobiota of marine macroalgae from the Tamilnadu coast: distribution, diversity and biotechnological potential. *Bot Mar*. 2010;53(5):457-468.
141. Collén JM, Mtolera M, Abrahamsson K, Semesi A, Pedersén M. Farming and physiology of the red algae *Eucheuma:* growing commercial importance in East Africa. *Ambio*. 1995;24(7–8):497-501.
142. Hurtado AQ, Neish IC, Critchley AT. Developments in production technology of *Kappaphycus* in the Philippines: more than four decades of farming. *J Appl Phycol*. 2015;27(5):1945-1961.
143. Ali MKM, Critchley AT, Hurtado AQ. Micropropagation and sea-based nursery growth of selected commercial *Kappaphycus* species in Penang, Malaysia. *J Appl Phycol*. 2020;32(2):1301-1309.
144. Tan J, Lim PE, Phang SM, et al. *Kappaphycus malesianus* sp. nov.: a new species of *Kappaphycus* (Gigartinales, Rhodophyta) from Southeast Asia. *J Appl Phycol*. 2014;26(2):1273-1285.
145. Lim PE, Tan J, Phang SM, et al. Genetic diversity of *Kappaphycus* Doty and *Eucheuma* J. Agardh (Solieraeae, Rhodophyta) in Southeast Asia. *J Appl Phycol*. 2014;26(2):1253-1272.
146. Halling C, Wikström SA, Lilliesköld-Sjöö G, Mörk E, Lundsør E, Zuccarello GC. Introduction of Asian strains and low genetic variation in farmed seaweeds: indications for new management practices. *J Appl Phycol*. 2013;25(1):89-95.
147. Liu N, Zhang L, Tang X, et al. Complete plastid genome of *Kappaphycus alvarezii*: insights of large-scale rearrangements among Florideophyceae plastid genomes. *J Appl Phycol*. 2019;31(6):3997-4005.
148. Tablizo FA, Lluisma AO. The mitochondrial genome of the red alga *Kappaphycus striatus* ("*Green Sacol*" variety): complete nucleotide sequence, genome structure and organisation, and comparative analysis. *Mar Genomics*. 2014;18(b):155-161.
149. Li Y, Liu N, Wang X, et al. Comparative genomics and systematics of *Betaphycus*, *Eucheuma*, and *Kappaphycus* (Silieriaceae: Rhodophyta) based on mitochondrial genome. *J Appl Phycol*. 2018;30(6):3435-3443.
150. Ritchie KB. Bacterial symbionts of Corals and *Symbiodinium*. In: Rosenberg E, Gophna U, eds. *Beneficial Microorganisms in Multicellular Life Forms*. Springer-Verlag; 2011:139-150.
151. Pirttilä AM. Commentary: Agroforestry leads to shifts within the gammaproteobacterial microbiome of banana plants cultivated in Central America. *Front Microbiol*. 2016;7:656.
152. Hurtado AQ, Reis RP, Loureiro RR, Critchley AT. *Kappaphycus* (Rhodophyta) cultivation: problems and the impacts of Acadian marine plant extract powder. In: Pereira L, Neto J, eds. *Marine Algae Biodiversity Taxonomy, Environmental Assessment and Biotechnology*. CRC Press; 2014:251-291.
153. Eggertsen M, Halling C. Knowledge gaps and management recommendations for future paths of sustainable seaweed farming in the Western Indian Ocean. *Ambio*. 2020;5:60-73.
154. Stuart EJ. Growing unculturable bacteria. *J Bacteriol*. 2012;194(16):4151-4160.
155. Xu HS, Roberts N, Singleton FL, Attwell RW, Grimes DJ, Colwell RR. Survival and viability of nonculturable *Escherichia coli* and *Vibrio cholerae* in the estuarine and marine environment. *Microb Ecol*. 1982;8(4):313-323.
156. Li L, Mendis N, Trigui H, Oliver JD, Faucher SP. The importance of the viable but non-culturable state in human bacterial pathogens. *Front Microbiol*. 2014;5:258.
157. Khan NH, Ahsan M, Taylor WD, Kogure K. Culturability and survival of marine, freshwater and clinical *Pseudomonas aeruginosa*. *Microbes Environ*. 2010;25(4):266-274.
158. Dale C, Moran NA. Molecular interactions between bacterial symbionts and their hosts. *Cell*. 2006;126(3):453-465.
159. Yang Q, Franco CMM, Zhang W. Uncovering the hidden marine sponge microbiome by applying a multi-primer approach. *Sci Rep*. 2019;9:6214.
160. Kemp KM, Westrich JR, Alabady MS, Edwards ML, Lipp EK. Abundance and multilocus sequence analysis of *Vibrio* bacteria associated with diseased Elkhorn coral (*Acropora palmata*) of the Florida Keys. *Appl Environ Microbiol*. 2018;84(2):e01035-e1117.
161. Dobretsov S, Véliz K, Romero MS, Tala F, Thiel M. Impact of UV radiation on the red seaweed *Gelidium lingulatum* and its associated bacteria [published online ahead of print July 2020]. *Eur J Phycol*. 2020. https://doi.org/10.1080/09670262.2020.1775309.
162. Campbell AH, Marzinelli EM, Gelber J, Steinberg PD. Spatial variability of microbial assemblages associated with a dominant habitat-forming seaweed. *Front Microbiol*. 2015;6:230.
163. Tan TT, Song SL, Poong S-W, Ward GM, Brodie J, Lim P-E. The effect of grazing on the microbiome of two commercially important agarophtes, *Gracilaria firma* and *G. salicornia* (Gracilariaceae, Rhodophyta). *J Appl Phycol*. 2020;32(4):2549-2559.
164. Xue M, Wu L, He Y, Liang H, Wen C. Biases during DNA extraction affect characterisation of the microbiota associated with larvae of the Pacific white shrimp, *Litopenaeus vannamei*. *PeerJ*. 2018;6:e5257.
165. Holt CC, Bass D, Stentiford GD, van der Geizen M Understanding the role of the shrimp gut microbiome in health and disease. [published online ahead of print, April 2020]. *J Inv Pathol*. 2020: https://doi.org/10.1016/j.jip.2020.107387.
166. García-López R, Cornejo-Granadas F, Lopez-Zavala AA, et al. Doing more with less: a comparison of 16S hypervariable regions in search of defining the shrimp microbiota. *Microorganisms*. 2020;8(1):134.
167. Douglas GM, Maffei VJ, Zaneveld JR, et al. PICRUSt2 for prediction of metagenome functions. *Nat Biotechnol*. 2020;38:669-688.
168. Aßhauer KP, Wemheuer B, Daniel R, Meinicke P. Tax4Fun: predicting functional profiles from metagenomic 16S rRNA data. *Bioinformatics*. 2015;31(17):2882-2884.
169. Bass D, Stentiford GD, Littlewood TD, Hartikainen H. Diverse applications of environmental DNA methods in parasitology. *Trends Parasitol*. 2015;31(10):499-513.
170. Zozaya-Valdes E, Roth-Schulze AJ, Egan S, Thomas T. Microbial community function in the bleaching disease of the marine macroalgae *Delisea pulchra*. *Environ Microbiol*. 2017;19(8):3012-3024.
171. Hudson J, Gardiner M, Deshpande N, Egan S. Transcriptional response of *Nautella italica* R11 towards its macroalgal host uncovers new mechanisms of host-pathogen interaction. *Mol Ecol*. 2017;27(8):1820-1832.
172. Longford SR, Campbell AH, Nielsen S, Case RJ, Kjelleberg S, Steinberg PD. Interactions within the microbiome alter microbial interactions with host chemical defences and affect disease in a marine holobiont. *Sci Rep*. 2019;9:363.
173. Work TM, Aeby GS. Pathology of tissue loss (white syndrome) in *Acropora* sp. corals from the Central Pacific. *J Inv Pathol*. 2011;107(2):127-131.
174. Séré MG, Tortosa P, Chabanet P, Quod J-P, Sweet MJ, Schleyer MH. Identification of a bacterial pathogen associated with Porites white patch syndrome in the Western Indian Ocean. *Mol Ecol*. 2015;24(17):4570-4781.
175. Sweet MJ, Burian A, Fifer J, Bulling M, Elliot D, Raymundo L. Compositional homogeneity in the pathobiome of a new, slow-spreading coral disease. *Microbiome*. 2019;7:139.
176. Bourne D, Garren M, Work TM, Rosenberg E, Smith GW, Drew HC. Microbial disease and the coral holobiont. *Trends Microbiol*. 2009;17(12):554-562.
177. Greene A, Leggat W, Donahue MJ, et al. Complementary sampling methods for coral histology, metabolomics and microbiome. *Methods Ecol Evol*. 2020;11:1012-1020.

178. Hayashi L, Santos AA, Faria GSM, et al. *Kappaphycus alvarezii* (Rhodophyta, Areschougiaceae) cultivated in subtropical waters in Southern Brazil. *J Appl Phycol*. 2011;23(3):337-343.
179. Pellizzari F, Reis RP. Seaweed cultivation on the Southern and Southeastern Brazilian coast. *Rev Bras Farm*. 2011;21(2):305-312.
180. Barbaroux O, Perez R, Dreno J-P. L'algue rouge *Eucheuma spinosum* possibilities d'exploitation et de culture aux Antilles (French language). *Science et Pêche Bull Inst Pêches Marit*. 1984;348.
181. Pang T, Liu Q. ENSO and eucheumatoid algae cultivation in China. *J Appl Phycol*. 2019;31(2):1207-1212.
182. Luxton DM, Robertson M, Kindley MJ. Farming of *Eucheuma* in the south Pacific Islands of Fiji. *Hydrobiologia*. 1987;151/152:359-362.
183. Arasamuthu A, Edward JKP. Occurrence of Ice-ice disease in seaweed *Kappaphycus alvarezii* at Gulf of Mannar and Palk Bay, Southeast India. *IJMS*. 2018;47(6):1208-1216.
184. Pambudi LT, Meinita MDN. Ariyati RW (2010) Seaweed cultivation in Indonesia: recent status. *J Mar Biosci Biotechnol*. 2010;4(1):6-10.
185. Handayani T, Alimuddin WU, Suryati E, Parenrengi A. Binary factor construction and *Agrobacterium tumefaciens*-mediated transformation of lysozyme gene in seaweed *Kappaphycus alvarezii*. *Biotropia*. 2014;21(2):80-90.
186. Achmad M, Alimuddin A, Widyastuti U, Sukenda S, Suryanti E, Harris E. Molecular identification of new bacterial causative agent of ice-ice disease on seaweed *Kappaphycus alvarezii*. *PeerJ Preprints*. 2016. https://doi.org/10.7287/peerj.preprints.2016v1 (not peer reviewed).
187. Mulyati H, Geldermann J. Managing risks in the Indonesian seaweed supply chain. *Clean Tech Environ Policy*. 2017;19(1):175-189.
188. Kimathi AG, Wakibia JG, Gichua MK. Growth rates of *Eucheuma denticulatum* and *Kappaphycus alvarezii* (Rhodophyta; Gigartinales) cultured using modified off-bottom and floating raft techniques on the Kenyan coast. *WIO J Mar Sci*. 2018;17(2):11-24.
189. Sulu R, Kumar L, Hay C, Pickering T. *Kappaphycus Seaweed in the Pacific: Review of Introductions and Field Testing Proposed Quarantine Protocols*. Secretariat of the Pacific Community. Aquaculture Technical Papers 2004: ISSN 1683-7568. (not peer reviewed).

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